

Highlights

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

Yunus Emre Cogurcu, Emrehan Metin

- Pre-fire thermal predictors add +0.06 to +0.15 AUC within all five fire regions
- The same block adds +0.004 AUC across 20 transfer directions, interval spans zero
- Highest niche-overlap pair fails both ways; lowest transfers both ways (points)
- Of 20 diagnostics, only conditional direction agreement tracks transfer (rho 0.84)
- Label-blind adaptation compresses transfer toward chance; only target labels help

Local skill, unstable portability: sign reversal in pre-fire thermal predictors across Mediterranean wildfire regions, and the diagnostics that do not order it

Yunus Emre Cogurcu^{a,*}, Emrehan Metin^a

^a*Çukurova University, Adana, Türkiye*

Abstract

Pre-fire thermal dryness separates a fire year from a normal year. Models built on it are rarely tested outside the region they were fitted in. Five Mediterranean wildfire regions were analysed here, on about 500 m cells, with MCD64A1 labels and spatially blocked validation. Six pre-fire thermal predictors were added to a terrain, fuel and greenness baseline.

The local skill does not travel, and neither does the baseline. Within-region ROC-AUC rose by +0.056 to +0.153 in every region. Across twenty ordered transfer directions the same block contributed +0.004 [−0.028, +0.036]. That is not distinguishable from zero, and its sign changes from pair to pair. The static baseline transferred no better, at 0.537 against 0.541. Two predictors reverse their association between regions: elevation and the LST anomaly. Removing them cost −0.081 of within-region skill, supported in every region and mostly due to elevation. It changed transfer by +0.014 [−0.017, +0.045]. Both figures are post-selection estimates. A local cost is measured. No compensating transfer gain is.

*Corresponding author.

Email address: `ycogurcu@cu.edu.tr` (Yunus Emre Cogurcu)

The failure is invisible to the diagnostics that can be run before deployment. Twenty candidate diagnostics were rank-correlated against observed transfer. Only two had intervals excluding zero, and both were conditional. The stronger one measures agreement in the sign of each predictor’s association (Spearman $\rho = 0.84$ over sixteen directions). No marginal measure was shown to order the matrix, including predictor-space dissimilarity. At the point estimates, the pair with the highest burned-niche overlap failed in both directions, while the lowest transferred in both. Signed associations need labels on both sides. The family that works is therefore the one a practitioner does not have.

The mechanism is a reversal of sign. Predictors do not weaken across regions. They reverse the direction of their association with burning. A distance in predictor space cannot see that. The same reversal appears inside one study area, between two fires eleven months apart on an identical grid. That arm rests on one fire and eleven positive-carrying blocks, so it corroborates rather than establishes.

Label-free alignment pushed fourteen of twenty directions towards chance rather than repairing them. Transfer skill has to be measured, not inferred from similarity. The price is target labels: thirty-two labelled 5 km blocks recovered 85 to 89 % of the target’s matched ceiling in three of six directions.

Keywords: wildfire, model transferability, area of applicability, land surface temperature, domain adaptation, concept shift

1. Introduction

Wildfire is a defining disturbance of Mediterranean-basin landscapes, and a changing climate is reshaping where and how it burns [27]. The pressure to anticipate *where* fire will occur has produced a large and still-growing literature on fire susceptibility mapping. It has recently been synthesised from both a methodological and an application perspective [39, 13]. The dominant pattern is well established. Geospatial predictors are assembled over a study region and paired with a historical record of burned area. A supervised classifier is then fitted, most often a random forest [4]. The resulting susceptibility surface is published with a cross-validated AUC in the 0.85 to 0.95 range.

This paper is about a property that the pattern does not report. Predictors that describe the *state* of a surface in a particular season are more informative about that season than predictors that describe a place. They are also, as we show, not portable: the relationship they carry is reparameterised locally, existing everywhere but pointing in different directions in different places. The dynamic block is worth +0.056 to +0.153 ROC-AUC inside every region and contributes +0.004 between them, an estimate whose interval spans zero, with a sign that is a property of the source-target pair rather than of the block.

The obvious reading of that pair of numbers is a trade-off, and our own earlier framing took it. The data do not support it. The static baseline transfers at a mean of 0.537 against the thermal model’s 0.541, so the predictor class that ought to travel does not travel either, and removing the reversing predictors returns +0.014 of transfer on an interval that also spans zero. Two

nulls on the portability axis are not an exchange. What is measured is a local gain that does not export, and a local cost when the reversing predictors are removed, with no compensating transfer gain on either arm.

1.1. Portability goes unmeasured

30 A susceptibility model’s reported skill is almost always an estimate of *within-region* performance. Held-out folds come from the same study area and season, often from the same fire event, and where folds are drawn at random over cells, spatial autocorrelation inflates the estimate further. The problem is documented across ecological modelling [31, 29] and is addressed
35 by spatially blocked cross-validation [37, 23]. Blocking makes the within-region estimate honest. It says nothing about whether the fitted relationship holds elsewhere.

Because only that side of the ledger is reported, the trade-off is structurally invisible. A predictor block is adopted on the strength of the in-
40 crement it delivers inside its training footprint, and the portability it consumes is never entered as a debit, even though any regional product built from locally trained models implicitly promises generalisation beyond that footprint. For fire specifically the transfer question is asked far less often than in adjacent fields. Meteorological fire-danger indices are known not
45 to port cleanly between fire environments [30], and the two recent studies that test model transfer systematically, across Mediterranean countries [7] and US counties [15], both report that it largely succeeds between similar regions. Both transfer models whose dominant predictors are *spatially stationary*: terrain, human modification, night-time lights, population density,
50 long-term climatologies. Each describes a place rather than a season, which is

the class our thesis expects to travel well. Species distribution modelling has examined transferability far more systematically [43], and there geographic and environmental similarity do *not* reliably predict transfer success [38, 32].

1.2. Pre-fire thermal dryness is the natural test case

55 The dynamic class used here is pre-fire thermal dryness, and it is the class most plausibly *expected* to transfer. Standard susceptibility predictors are terrain, fuel type, greenness, long-term climatology and proximity to roads and settlements, and most are static or near-static over the timescale at which fire danger varies. Such a set explains poorly why a particular summer
60 burned and the preceding one did not. What changes between those summers is the state of the surface: fuel dryness, moisture stress, and how anomalously hot the land surface has become relative to its own baseline. Satellite thermal observation gives direct access to part of that state through fuel moisture content [44], and pairing surface temperature with a vegetation index was
65 established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [6]. The Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure which should, in principle, be less exposed to absolute-temperature offsets between regions than raw land surface temperature. That theoretical portability advantage
70 is tested here.

That pre-fire thermal state carries genuine information about subsequent fire is established [17, 19, 18, 8]. What has not been established is whether the predictive *skill* of a classifier trained on it in one fire region survives strict, label-free application to another. The physics linking moisture stress
75 to combustion is universal, so portability should be most expected for this

class, which is what makes it diagnostic: a loss here cannot be dismissed as a peculiarity of a locally defined covariate.

1.3. Why the loss is invisible to the diagnostics in use

A deficit on transfer can arise two ways. Under **covariate shift** the
80 marginal distribution of the predictors differs while the predictor-response
relationship is preserved, which is in principle correctable without target labels, by per-region standardisation or covariance alignment such as CORAL [35], an approach with an established remote-sensing literature [36]. Under **concept shift** the conditional relationship itself changes: a predictor positively associated with burning in one region is negatively associated with
85 it in another. No label-free realignment of input distributions can repair a sign reversal, because the information needed to detect it is exactly the information being withheld. The taxonomy is canonical [25].

This distinction is what makes the failure hard to anticipate. The diagnostics the field relies on to decide whether a model may be applied to
90 new ground, above all area-of-applicability measures built on dissimilarity in predictor space [21, 22, 16], are marginal by construction. A target region can sit well inside the training data’s predictor envelope while the relationship between those predictors and burning points the other way. This paper
95 measures both sides on the same pairs.

1.4. Contributions

Three findings carry this paper. Each is stated at the strength its interval supports, and the nearest prior work is named inside the claim.

Contribution 1. Local skill and portability, measured separately
 100 **and per direction.** The thermal block is worth +0.056 to +0.153 ROC-AUC inside every one of five regions, with every bootstrap interval above zero and the result holding in both analysis populations. Across the twenty ordered transfer directions it contributes +0.004. That interval spans zero under all four between-direction resampling units we computed, and the sign
 105 varies by pair: paired deltas run from -0.148 to $+0.132$, twelve positive and eight negative. The baseline arm bounds the reading. It transfers at a mean of 0.537, against the thermal model’s 0.541. The static predictor class is not the portable one here either. What the matrix shows is that **nothing in this feature space travels**, not that dynamic state is uniquely fragile. Feature
 110 removal measures the local cost. Dropping the two reversing predictors, elevation and the LST anomaly, costs -0.081 of mean within-region AUC, supported in every region. Roughly three quarters of that is elevation, a *baseline* terrain variable. Mean transfer changes by +0.014, whose interval also spans zero. Both are post-selection estimates (Section 4.6b). A local cost is mea-
 115 sured. No compensating transfer gain is. The within-region half is not novel, comparable results existing for these landscapes [1, 12]; the paired contrast against portability is. Dimarco et al. [7] transfer a predominantly stationary predictor set successfully across a comparable Mediterranean design, though their response variable is ignition rather than burned area (Section 5.7).

120 **Contribution 2. The loss is invisible to the diagnostics that could be run before deployment, and visible to one that cannot.** Twenty candidate diagnostics from four families are evaluated against observed transfer under one bootstrap framework. Area-of-applicability-style

predictor-space dissimilarity [21, 22, 16], climatic and geographic distance,
 125 learned domain separability, niche-overlap statistics and burn-pattern regime
 distances were none of them shown to order the matrix; the domain classifier
 is at ceiling, with $AUC \geq 0.96$ for every pair. Only two diagnostics have in-
 tervals excluding zero. Both measure conditional direction agreement, that
 is, whether each predictor’s signed association with burning points the same
 130 way in both regions (Spearman $\rho = +0.84$ and $+0.81$ over sixteen directions
 from eight pairs). Both are computed on a data-selected subset of predic-
 tors, and their all-nine-feature counterparts span zero, so the result rests
 on that selection. Equalising the families onto a common twelve directions
 leaves the ordering unchanged, so it is not an artefact of unequal samples.
 135 Signed associations need burned labels on both sides, so this is a mechanism
 diagnosis rather than a pre-deployment screen, and the marginal family, the
 only one runnable before deployment, is the one that fails. The same point
 survives without any ranking, at the point estimates: the pair with the high-
 est burned-niche overlap fails in both directions while the lowest transfers
 140 in both. That contradicts an expectation the fire literature carries [7, 15]
 and agrees with species distribution modelling [38, 32]. Shift decomposition
 in applied remote sensing is not itself new [11]; the fire application and the
 head-to-head of four diagnostic families on the same pairs are.

Contribution 3. The mechanism is a reversal of sign, and it sur-
 145 **vives holding geography fixed.** Predictors do not merely weaken across
 regions, they reverse the direction of their association with burning, which
 is why a distance in predictor space cannot see the failure. The sharpest
 reversal is elevation, whose association points opposite ways in Manavgat

and in Bejís and Muğla with disjoint bootstrap intervals. The same reversal
150 appears inside a single study area, between two fires eleven months apart
on an identical grid, where season and population also differ but place does
not (Section 4.8). That arm rests on one fire and eleven positive-carrying 5
km blocks, below the floor this design sets for itself, so it corroborates the
mechanism rather than establishing it.

155 Two consequences follow. Both are reported as supporting results, not
leading ones. First, label-free alignment by standardisation and covariance
alignment [35] does not repair transfer. It compresses transfer towards chance
in fourteen of twenty directions. We believe this is the first application
of covariance alignment to fire susceptibility. Second, the residual gap is
160 conditional, so the resource that closes it is target labels. That price is
measurable. Thirty-two labelled 5 km blocks recover 85 to 89 % of the
target’s matched ceiling in three of six directions, 51 to 57 % in two more,
and 30 % in the sixth.

Alongside these, a leakage-audited, spatially blocked evaluation and trans-
165 fer protocol is released with code, configuration and frozen outputs, so that
most of this result can be re-run rather than taken on trust. One component
is an exception and is named in the data-and-code availability statement:
the reproduction-check code is not in the released repository at the commit
of record. The few-shot export, previously also listed as an exception, was
170 regenerated at commit `6d7a6a71` and its validator now passes on every check.
This matters given evidence that wildfire transfer conclusions are sensitive
to evaluation design and task formulation [42].

A companion paper treats the observational layer beneath this one, where

preprocessing decisions taken before any model is fitted are shown to carry
175 their own uncertainty budget.

Section 2 reviews the relevant literature. Section 3 describes the study regions, the data, and the modelling, cross-validation, transfer and adaptation protocol. Section 4 reports results. Section 5 discusses them, including limitations. Section 6 concludes.

180 2. Related work

2.1. Fire susceptibility mapping, and what it reports

Fire susceptibility mapping has been synthesised recently from both methodological and application perspectives [39, 13]. The dominant pattern is stable: geospatial predictors are assembled over a study region, paired with a
185 historical burned-area record, and fitted with a supervised classifier, most often a random forest [4], and the resulting surface is published with a cross-validated AUC in the 0.85 to 0.95 range. That figure is a within-region estimate. Comparable within-region results already exist for the landscapes studied here [1, 12].

190 *2.2. Pre-fire thermal dryness*

Satellite thermal observation gives repeated access to surface state through fuel moisture content [44]. Pairing land surface temperature with a vegetation index was established as a live-fuel-moisture estimator for fire-danger rating by Chuvieco et al. [6], and the Temperature-Vegetation Dryness Index [33] formalises that feature space into an internally normalised measure,
195 which is why it is often expected to travel between regions better than raw

temperature. That pre-fire thermal state carries information about subsequent fire is established [17, 19, 18]. Gelabert et al. [8] found dead fine fuel moisture and its anomalies to be the most influential predictor of human-caused ignition likelihood across Europe, testing generalisation by pooled fitting with per-site evaluation. What has not been tested is whether a classifier built on this class survives strict, label-free application to a region it never saw.

2.3. Spatial validation and transferability

Random cross-validation over spatially autocorrelated cells inflates skill estimates [31, 29], and spatially blocked designs are the standard remedy [37, 23]. The remedy is not uncontested: blocked schemes have been argued to introduce pessimistic bias of their own [40, 24, 5]. This paper uses spatial blocking throughout and reports every result at three block sizes, so that a reader can see which verdicts depend on the choice.

Blocking makes a within-region estimate honest but says nothing about another region. For that, the field’s instrument is the area of applicability and related predictor-space dissimilarity measures [21, 22, 16], which ask whether a target’s predictor values fall inside the training data’s envelope. Species distribution modelling has examined transferability far more systematically than fire has [43], and two studies there report that geographic and environmental similarity do not reliably predict transfer success [38, 32], and the transferability of satellite-derived empirical models is an active question in fire remote sensing itself [20].

220 *2.4. Dataset shift and label-free adaptation*

The canonical taxonomy separates covariate shift, where the predictor distribution moves but the predictor-response relationship holds, from concept shift, where the relationship itself changes [25]. Covariate shift is in principle correctable without target labels, by per-region standardisation or
225 by covariance alignment such as CORAL [35], and domain adaptation has an established remote-sensing literature [36, 28]. Concept shift is not: no realignment of input distributions can repair a reversal in the sign of an association, because detecting it requires the labels being withheld. Shift decomposition in applied remote sensing is not itself new [11]. We are not
230 aware of a prior application of covariance alignment to fire susceptibility, fire occurrence or burned-area prediction.

2.5. Cross-region generalisation of fire models

Few studies test fire-model transfer directly. Podschwit et al. [30] give the long-standing reference point that meteorologically derived danger indices do not port cleanly between fire environments. WildfireGenome [15]
235 runs a leave-one-county-out matrix across seven US counties and reports strong within-county performance with highly variable off-diagonal transfer, similar pairs transferring well and dissimilar pairs collapsing; its label, however, is a principal-component composite of hazard *indicators* rather than observed burned area, and it applies no adaptation. Xu et al. [42] argue that
240 wildfire transfer conclusions depend strongly on evaluation design, a caution we address by fixing the protocol in a project log before the diagnostics were computed and by reporting every sensitivity axis; that log is not a formal pre-registration, and Section 3.11 states what was fixed and when. Kondylatos

245 et al. [14] provide Mesogeos, a 1 km Mediterranean datacube for data-driven wildfire modelling.

The nearest neighbour, and the contrast the present paper draws.

Dimarco et al. [7] is the closest Mediterranean analogue: 500 m predictors harmonised across four countries, tree ensembles under spatial cross-validation, transfer tested both leave-one-country-out and as a full 4 x 4
250 matrix, with every transfer exceeding AUC 0.80 and bioclimatically similar countries scoring higher. They apply no domain adaptation.¹

Much of the design is shared — Mediterranean regions, 500 m cells, MCD64A1-derived targets, tree ensembles, spatially aware validation, an ex-
255 plicit transfer matrix — and two things differ. **The predictor class:** their model rests on attributes of a place, all spatially stationary, ours on the state of a surface in one season. **The response variable:** theirs is human-driven ignition, ours is burned area, and those are different quantities with different dominant controls. Read together the two results suggest that the relation-
260 ship between domain similarity and transfer success may be predictor-class dependent, with Vesk et al. [38] and Rousseau and Betts [32] aligning from species distribution modelling while Dimarco et al. and WildfireGenome run against it. We present that as a live disagreement, and Section 5.7 states why our own data do not settle it.

265 Methods, where no land surface temperature or TVDI variable appears and the only temperature predictor is a static ERA5-Land seasonal climatology [26]. We follow their Methods, which we take to be authoritative, and

¹Their Results text refers to "LST anomalies", which is inconsistent with their own

note the discrepancy so that the comparison is transparent. Every statement about their work was checked against the full text, read 2026-08-08.

270 3. Data and methods

3.1. Study regions and temporal windows

Five Mediterranean wildfire regions are analysed: Manavgat 2021 and Muğla 2021 in Türkiye, Bejís 2022 in Spain, North Evia 2021 in Greece and Montiferru 2021 in Sardinia. Each is a place-based rectangular area of
275 interest in EPSG:4326, defined from place coverage rather than from a fire perimeter, and not tuned on burned prevalence, gate outcome or any model metric. One AOI choice was label-informed and is stated as such. The North Evia box was extended after the legacy box was found to carry an atypically high burned prevalence. The extended geometry was then defined from place
280 anchors, and the legacy variant is kept as a sensitivity arm (Appendix A). The AOIs are deliberately not clipped to fire perimeters, so that unburned cells around each fire form the negative class rather than being excluded by construction. A sixth region, Kozan 2023, is carried as a negative control and is excluded from all modelling by the gate of Section 3.3.

285 Each region has two non-overlapping windows. The **predictor window** closes the day before the **label window** opens, so no predictor observation can post-date the first labelled burning. Window lengths are matched within a region and vary between them with the event: 56 to 60 days for predictors and 35 to 49 days for labels. A four-year baseline of window-symmetric
290 composites precedes each predictor window and supplies the climatological reference for the anomaly channels. Region identifiers, bounding boxes, win-

down dates and baseline years are registered in `core/regions.py` and are reproduced in Table 1.

3.2. Burned-area label and the ~ 500 m analysis grid

295 Labels come from MODIS MCD64A1 Collection 6.1 [9] retrieved through Google Earth Engine [10], whose omission and commission characteristics [3] bound every model here. The analysis grid is **reconstructed rather than native**: the pipeline’s 30 m EPSG:4326 reference grid is partitioned into 17 x 17 blocks, giving a nominal 510 m cell that approximates rather
300 than reproduces the MODIS cell, and is square in degrees but not on the ground. A cell’s burn date is the mode of its positive sub-pixel day-of-year values, tested against the region’s label window; the label never affects a cell’s eligibility for modelling. Two safeguards are recorded rather than assumed: cells burning inside a region’s own predictor window are removed, which ran
305 for three of five regions, and burning in earlier years is screened for none. Appendix C.1 gives the full specification, including what follows from the grid’s shape.

3.3. Burned-landcover admissibility gate

 Before any modelling each region passes a gate asking what fraction of
310 its burned cells is dominated by natural vegetation, using ESA WorldCover classes on the same cells. A region is admitted at 0.50 with at least 30 burned cells, and rejected as a cropland-dominated control at 0.50 cropland. The purpose is to separate burned area produced by natural-fuel combustion from post-harvest stubble burning, which MCD64A1 does not distinguish.
315 Verdicts are in Section 4.1; the full rule is in Appendix C.1.

3.4. Predictor variables

Ten predictors are used, four baseline and six thermal, all summarised per cell over the predictor window. All optical and thermal predictors come from Landsat 8 Collection 2 Level-2 (LANDSAT/LC08/C02/T1_L2), quality-screened per pixel from the QA_PIXEL band with the water bit deliberately preserved; the coarse-resolution thermal input is MODIS/061/MOD11A1.

- **NDVI**, the predictor-window median of Landsat surface reflectance, is the baseline’s one time-varying member.
- **Elevation** and **slope** come from the Copernicus DEM GLO-30.
- **Land cover** is ESA WorldCover v200 [45], entering as the dominant class code.
- **Current LST**, the predictor-window median of Landsat surface temperature in °C.
- **LST anomaly**, a z-score of the current-window LST median against the four baseline years, set to no-data where the baseline standard deviation is below 1.0 °C or observations are too few.
- **TVDI** and **TVDI difference**. The Temperature-Vegetation Dryness Index [33] is computed in the LST-NDVI feature space, with wet and dry edges taken as the 2nd and 98th LST percentiles within each of twenty NDVI bins and the index clamped to [0, 1]. The edges are percentiles of the LST values a given scene contains, so they are fitted per AOI and per window, and a TVDI of 0.5 denotes a different

physical moisture state in each region. The companion paper reports what that scene dependence does and does not explain. **TVDI difference** is the raw anomaly of that index against the same four window-symmetric baseline years used for the LST anomaly: `tvdi_difference = current_tvdi - mean(baseline_tvdi)`, in index units rather than standard deviations, which is why it is reported alongside the z-scored channel rather than in place of it.

- **Downscaled LST and fused LST.** A MODIS-to-Landsat downscaling model is trained on the predictor window and applied to the full 30 m grid. The fused product equals observed Landsat LST wherever that is valid, and the downscaled surface only where it is not. Gap-filling therefore never replaces or blends a valid observation. The gap-filled share is 0.11 % to 9.70 % by region. The downscaler’s own inputs include coordinates, which is the one route by which a coordinate-derived surface re-enters a feature set from which Section 3.13 excludes coordinates; the companion paper measures what the increment is worth without these two channels.

3.5. Cell aggregation, validity and analysis populations

Cell-level values are means over the ~ 510 m cell, computed from valid 30 m pixels only, with the valid fraction recorded. A cell is `valid_for_modeling` when it meets a 30 % valid-pixel floor and no thermal channel is missing. Two populations are then defined. The **primary** population is natural vegetation, cells whose combined tree, shrub and grass fraction reaches 0.50, which excludes cropland from every burnable mask. The **sec-**

ondary population is all valid cells, reported throughout as a sensitivity. Every headline result is given for both.

3.6. Feature sets and classifier

365 Two nested feature sets are compared. The **baseline** set is NDVI, elevation, slope and land cover. The **thermal** set is the baseline plus the six thermal channels. Land cover is one-hot encoded and numeric features are standardised inside the fitting pipeline. The classifier is a random forest [4] with 300 trees, unlimited depth, `min_samples_leaf = 3`, balanced
370 class weights and `random_state = 42`, identical for every region, population, feature set and transfer direction, so that no comparison in this paper is confounded by a model choice.

3.7. Spatial-block cross-validation and bootstrap uncertainty

Cross-validation is spatially blocked. Blocks are squares of the analysis
375 grid formed as `row_500m // B by col_500m // B`, and folds are drawn over whole blocks, so no block is split between training and test. Five folds are used. The block size B is reported at 2, 10 and 20 cells, about 1, 5 and 10 km, for the within-region analysis.

Uncertainty is a spatial-block bootstrap: blocks are resampled with replacement, 1000 replicates, seed 42, with 2.5 and 97.5 percentiles as the
380 interval. Because it is blocks that are resampled, what bounds an interval's reliability is the number of blocks carrying at least one burned cell, and those counts are reported alongside the intervals. Where a verdict rests on too few such blocks it is stated as indicative rather than as an interval.

385 3.8. Cross-region transfer protocol

For each ordered pair of regions the model is fitted on the source population and applied to the target population with **no target label used at any point**: no refitting, no threshold selection, no calibration. Both the baseline and the thermal feature set are transferred, so that the thermal
 390 block’s contribution is a paired difference within a direction rather than a comparison across directions. All twenty ordered directions among the five regions are computed.

3.9. Label-blind domain adaptation

Two label-free remedies are tested on every direction, in both feature sets.
 395 **Region-wise z-score.** Each region’s numeric features are standardised using its own statistics, source statistics from source data and target statistics from target data, never pooled. The classifier is refitted on the z-scored source and applied to the z-scored target. This removes first- and second-order marginal offsets.

400 **CORAL after region-wise z-score.** Starting from the z-scored features, the source covariance is aligned to the target covariance by the standard whitening-recolouring map [35], with regularisation $\lambda = 10^{-5}$ and eigenvalues floored at 10^{-12} . Critically **the transform is applied to the source only**; the target is left exactly as it is. The classifier is refitted on the aligned
 405 source. Neither variant sees a target label, and both are verified label-blind at run time.

λ sensitivity was assessed over nine values from 0 to 10^{-1} on four directions in both feature families, moving transfer AUC by at most 0.014 in

any direction and 0.008 within the thermal family. The canonical $\lambda = 1$ lies
 410 outside the released sweep; the companion paper reports it.

3.10. *Transfer-gap decomposition and the concept-shift diagnostic*

For each direction the gap between the target’s own within-region skill
 and the raw transfer result is split in two. One part is what the best label-free
 adaptation recovers, and the other is what it does not. The recovered fraction
 415 is defined as (adapted – raw) / (within – raw), signed and unclipped, and its
 interval comes from the same paired bootstrap. The recovered part bounds
 what covariate-level correction can achieve. The remainder is the conditional
 residual.

The mechanism is diagnosed by **signed univariate association**. For
 420 each numeric predictor the raw ROC-AUC of that predictor against **burned**
 is computed in each region and never folded to $\max(\text{AUC}, 1 - \text{AUC})$, so a
 value below 0.5 is read as a direction rather than as weakness. A reversal is
 called bootstrap-supported only when the two regions’ intervals are disjoint,
 under the same 10-cell spatial-block bootstrap.

425 3.11. *Transferability diagnostics versus transfer*

Twenty candidate diagnostics from four families are computed for every
 region pair and rank-correlated (Spearman) against observed raw transfer
 AUC, under one bootstrap framework that resamples unordered region pairs
 with both of their ordered directions travelling together. The families are
 430 marginal predictor-space measures $P(x)$, including area-of-applicability dis-
 similarity, climatic and geographic distance and a learned domain classifier;

burned-niche overlap $P(x|y=1)$; burn-pattern regime distance $P(y)$; and conditional direction agreement $P(y|x)$. No family-wise error control is claimed, and the number of variants computed within each family is reported with
435 the result. Appendix C.1 gives the full specification of each diagnostic and of the bootstrap.

3.12. Interventions

Pooled multi-region training. Leave-one-region-out: the model is trained on the pooled primary populations of the other four regions and
440 evaluated on the held-out region, with folds blocked as before. This asks whether pooling recovers what single-source transfer loses.

Removal of direction-reversing features. The two predictors whose signed association reverses between regions with bootstrap support are **elevation_mean** (Manavgat against Bejís and against Muğla) and **lst_anomaly_mean**
445 (Bejís against Evia). They are dropped and everything is refitted, within-region and across every direction, and the two are also dropped singly so the cost can be attributed. This measures what the removal costs locally and what it returns on transfer on the same footing. Note that the two features are selected by the same reversal analysis against which the result is then
450 read, so both quantities are post-selection estimates and no correction for that selection is applied.

3.13. Leakage control

An explicit forbidden-column set is enforced at every model fit. Coordinates (**lon**, **lat**, **row**, **col** and their normalised forms), every burn-date and
455 label-provenance column, and the agreement fraction are excluded from all

feature sets, and the enforcement runs as an assertion rather than a convention. The natural-vegetation mask is used only to define the population, never as a predictor. Spatial blocking prevents a cell from sharing a fold with its own neighbours.

460 **Reproducibility.** All randomness uses seed 42 and the bootstrap uses 1000 replicates throughout. The transfer and adaptation analysis runs in an environment separate from the upstream pipeline’s. Every within-region model was therefore refitted there and compared against the frozen upstream output, and the independently implemented adaptation was compared
465 against the pipeline’s own. The within-region comparisons agree exactly, and the twenty transfer directions agree to within 1.6×10^{-7} . Random-forest fits are not bit-identical across library versions. All numbers here were therefore produced under scikit-learn 1.9.0, or verified against it. If that version is not fixed, cross-region point estimates carry an implementation
470 tolerance of about ± 0.02 to 0.03. The companion paper reports the version sensitivity and the reproduction check in full.

Sensitivity analyses. Every headline result is repeated across two analysis populations, three spatial-block sizes, the CORAL regularisation sweep and both feature sets. Where a conclusion depends on one of those choices,
475 the dependence is reported rather than resolved by choosing the favourable setting.

3.14. Same-geography event-to-event comparison (Mugla 2021 versus 2022)

 Mugla admits a comparison in which place is held fixed and the event varies: a second fire burned inside the identical AOI, on the identical analysis
480 grid, eleven months after the first. Signed univariate AUCs are computed

for both arms under the same 10-cell spatial-block bootstrap used elsewhere. Season, year and population all differ between the arms, since the 2022 arm is defined by removing the 2021 scar, and Section 4.8 reports what that costs. Appendix C.2 gives the window dates, the population construction and the
485 asymmetry audit.

4. Results

4.1. Study regions and the admissibility gate

All five candidate regions pass the burned-landcover gate as wildfire candidates and the negative control fails it as intended. Kozan 2023 returns
490 a natural-vegetation fraction of 0.017 against the 0.50 threshold and is excluded from all modelling. The separation is not marginal: the five admitted regions carry fractions of 0.723 to 0.991, so the threshold falls in an empty interval rather than between neighbouring cases. That is consistent with the gate separating burned area produced by natural-fuel combustion from post-
495 harvest stubble burning, which MCD64A1 does not distinguish, though one negative control cannot establish it.

North Evia is analysed on an extended AOI. Relative to the legacy box, the extended box leaves the burned scar essentially unchanged while cutting overall prevalence from 0.361 to 0.122 and TSG prevalence from 0.676 to
500 0.287. The effect of that choice on transfer is reported in Appendix A(a).

4.2. Within-region: the thermal increment replicates in five regions

Adding the six thermal predictors to the baseline raises spatially blocked out-of-fold ROC-AUC in every region. The increment's bootstrap interval

Table 1: Region summary (recap of Table 1; final numbering at assembly). Counts from each region’s Step 8A dataset statistics; gate fractions from each region’s burned-landcover gate output. TSG = the primary natural-vegetation population. The TSG columns use the canonical modelled population, **burnable_tree_shrub_grass and valid_for_modeling == True**, which is the population every model in this paper was fitted and scored on.

Region	Total cells	Valid cells	Burned	Prevalence (all valid)	TSG cells	Burned in TSG	TSG prevalence	Burned natural- veg frac- tion	Gate verdict
Manavgat 2021	24,150	24,087	796	0.033	20,511	784	0.038	0.984	pass
Bejis 2022	15,759	15,759	1,103	0.070	15,190	1,100	0.072	0.991	pass
Muğla 2021	73,098	73,045	3,026	0.041	41,730	2,911	0.070	0.958	pass
North Evia 2021 (ex- tended)	22,925	22,906	2,788	0.122	9,298	2,664	0.287	0.945	pass
Monticelli 2021	3,234	3,173	697	0.220	2,544	539	0.212	0.723	pass

excludes zero in all five regions at 1 km and at 5 km blocking. Those are the
 505 two scales this design supports as intervals. At 10 km the point estimates
 hold, from +0.048 to +0.154. They rest on 6 to 33 positive-carrying blocks,
 so they are reported as indicative rather than as intervals. The table note
 gives the reason.

Table note (resampling units). The bootstrap resamples spatial blocks,
 510 so what bounds an interval’s reliability is the number of blocks that carry
 at least one burned cell. Those counts fall quickly as blocks coarsen. At
 2 cells they are 235 (Manavgat), 302 (Bejís), 843 (Muğla), 716 (Evia) and
 192 (Montiferru), out of 5 439, 3 967, 11 316, 2 566 and 743 blocks. At 10
 cells they are 28, 19, 70, 41 and 16, out of 237, 176, 576, 155 and 35. At
 515 20 cells they are **12, 6, 33, 15 and 6**, out of 60, 48, 167, 50 and 12. No
 bootstrap replicate was invalid at any block size, with one exception. At
 20 cells Bejís had 5 of 1000 replicates single-class. That is the symptom
 of the six positive-carrying blocks just reported, and a further reason to
 read the row as indicative. An equal-tailed percentile interval built on six
 520 positive-carrying blocks has no meaningful coverage. Montiferru at 20 cells
 also feeds only 12 groups into a 5-fold grouped split, so its models train on
 about ten blocks each. **The 20-cell row of this table should be read
 as indicative rather than as an interval.** The 10-cell row is the coarsest
 blocking this design supports properly, because every region there has 16
 525 to 70 positive-carrying blocks. The increment holds at that scale in all five
 regions. Source: `paper/referee2_numbers.md`, block C, counted from the
 frozen per-cell prediction tables.

This is the first half of the trade-off. It is not itself novel, and it is reported

Table 2: Within-region baseline versus thermal performance and block-size robustness. Primary (TSG) population; spatially blocked 5-fold CV (Section 3.8); paired spatial-block bootstrap, 1000 replicates. Block sizes 2/10/20 cells $\approx$ 1/5/10 km.

Region	Block ($\approx$ scale)	Baseline AUC	Thermal AUC	Δ AUC	Δ AUC 95% CI
Manavgat 2021	2 ($\sim$ 1 km)	0.803	0.870	+0.067	[+0.055, +0.079]
	10 ($\sim$ 5 km)	0.748	0.797	+0.050	[+0.023, +0.077]
	20 ($\sim$ 10 km)	0.683	0.731	+0.048	[+0.014, +0.085]
Bejís 2022	2 ($\sim$ 1 km)	0.862	0.918	+0.056	[+0.048, +0.065]
	10 ($\sim$ 5 km)	0.779	0.825	+0.045	[+0.018, +0.069]
	20 ($\sim$ 10 km)	0.739	0.795	+0.057	[+0.031, +0.090]
Muğla 2021	2 ($\sim$ 1 km)	0.743	0.859	+0.116	[+0.106, +0.125]
	10 ($\sim$ 5 km)	0.698	0.777	+0.079	[+0.050, +0.105]
	20 ($\sim$ 10 km)	0.673	0.733	+0.061	[+0.030, +0.094]
North Evia 2021 (ext.)	2 ($\sim$ 1 km)	0.759	0.912	+0.153	[+0.142, +0.166]
	10 ($\sim$ 5 km)	0.716	0.864	+0.148	[+0.119, +0.183]
	20 ($\sim$ 10 km)	0.679	0.833	+0.154	[+0.124, +0.189]
Montiferu 2021	2 ($\sim$ 1 km)	0.781	0.883	+0.101	[+0.080, +0.125]
	10 ($\sim$ 5 km)	0.620	0.720	+0.100	[+0.017, +0.186]
	20 ($\sim$ 10 km)	0.555	0.681	+0.126	[+0.053, +0.228]

because the second half is measured against it.

530 *4.3. Cross-region transfer, and what label-free adaptation does to it*

Raw transfer is heterogeneous and includes anti-predictive directions. Raw target AUC spans 0.326 to 0.686. Twelve of 20 directions are above chance with CI support. Six are *below* chance with CI support: both directions of Manavgat to Bejís and of Manavgat to Muğla, plus Bejís to Evia and Evia to Bejís. Two intervals span 0.5. Even the best raw transfer,
535 Evia to Manavgat at 0.686, stays far below that target’s own within-region thermal performance of 0.870. Across all directions the raw deficit against the within-region reference is 0.184 to 0.592 AUC.

Those counts belong to the 2-cell blocking of Table 4. At the more conservative
540 10-cell blocking the same points give 9 above, 4 below and 7 uncertain, and no direction changes side of the chance line (Section 4.7). Four of the six below-chance directions keep their support there. Bejís to Manavgat and Manavgat to Muğla lose it and carry no verdict. The qualitative statement is unchanged. The counts should not be read as exact.

The static baseline does not transfer either. This is the control
545 for the reading the rest of the paper invites, and it constrains that reading sharply. The same twenty directions were run with the terrain, fuel and greenness baseline alone. The mean target AUC is **0.537**, against **0.541** for the thermal model. Static attributes of a place are the class Dimarco
550 et al. transfer successfully, and are the class this paper’s framing treats as portable. Here that class is itself barely above chance. The paired per-direction contrast is given in Section 4.6b. What matters here is that the transfer failure below is not specific to the dynamic block. A baseline that

does not travel, plus pre-fire thermal state, gives a model that does not travel.

555 **The thermal block’s paired contribution to transfer, with its interval.** The two matrices were differenced direction by direction. The mean is **+0.004**. The directions are not independent, because each region appears in eight of the twenty. The interval therefore depends on what is treated as the resampling unit. All four units the design permits give the
560 same answer:

Table 3:

Resampling unit	n	95 % interval on the mean paired contribution
Directions, naive	20	[−0.027, +0.034]
Unordered pairs, cluster bootstrap	10	[−0.028, +0.036]
Unordered pairs, t on pair means	10	[−0.034, +0.042]
Regions, leave-one-out jackknife	5	[−0.037, +0.046]

Every interval spans zero, and the point estimate is a small fraction of each width. The per-region jackknife shows how little the mean is anchored. Holding out Manavgat, Bejís, Muğla, Evia and Montiferru in turn gives +0.0148, +0.0021, +0.0065, **−0.0081** and +0.0060. **Dropping Evia alone**
565 **reverses the sign of the headline.** None of these four units propagates within-direction sampling variability. They resample between directions only.

Label-blind adaptation compresses the matrix toward chance rather than repairing it. Under region-wise z-scoring the twenty directions span 0.431 to 0.630 and under CORAL 0.443 to 0.624, roughly half the
570 raw spread, with no adapted direction exceeding 0.631 against within-region references of 0.859 to 0.918. Adaptation raises the failing directions and

Table 4: Cross-region transfer matrix, thermal model, TSG population. Target ROC-AUC with 2-cell spatial-block bootstrap 95% CIs (1000 replicates). CORAL is applied after region-wise z-scoring ($\lambda = 10^{-5}$).

Direction	Raw	Region-wise z-score	CORAL
Manavgat→Bejís	0.326 [0.305, 0.349]	0.477 [0.451, 0.502]	0.511 [0.484, 0.534]
Bejís→Manavgat	0.444 [0.408, 0.480]	0.457 [0.420, 0.497]	0.555 [0.528, 0.583]
Manavgat→Muğla	0.470 [0.451, 0.488]	0.431 [0.411, 0.449]	0.443 [0.423, 0.461]
Muğla→Manavgat	0.401 [0.378, 0.426]	0.559 [0.531, 0.587]	0.560 [0.535, 0.587]
Manavgat→Evia	0.613 [0.593, 0.631]	0.542 [0.520, 0.565]	0.539 [0.518, 0.561]
Evia→Manavgat	0.686 [0.653, 0.716]	0.516 [0.489, 0.544]	0.527 [0.500, 0.553]
Bejís→Muğla	0.618 [0.601, 0.635]	0.518 [0.501, 0.535]	0.507 [0.489, 0.524]
Muğla→Bejís	0.583 [0.561, 0.607]	0.535 [0.512, 0.557]	0.560 [0.538, 0.581]
Bejís→Evia	0.383 [0.363, 0.402]	0.532 [0.509, 0.551]	0.499 [0.479, 0.518]
Evia→Bejís	0.448 [0.426, 0.470]	0.549 [0.524, 0.575]	0.549 [0.525, 0.573]
Muğla→Evia	0.653 [0.636, 0.671]	0.561 [0.543, 0.580]	0.563 [0.545, 0.582]
Evia→Muğla	0.577 [0.560, 0.593]	0.501 [0.485, 0.518]	0.530 [0.515, 0.546]
Montiferru→Manavgat	0.567 [0.539, 0.594]	0.573 [0.538, 0.609]	0.606 [0.574, 0.639]
Manavgat→Montiferru	0.533 [0.488, 0.580]	0.586 [0.540, 0.629]	0.592 [0.550, 0.631]
Montiferru→Bejís	0.548 [0.521, 0.578]	0.574 [0.552, 0.596]	0.569 [0.548, 0.591]
Bejís→Montiferru	0.594 [0.560, 0.631]	0.550 [0.500, 0.601]	0.574 [0.530, 0.621]
Montiferru→Muğla	0.619 [0.604, 0.634]	0.576 [0.562, 0.589]	0.565 [0.550, 0.579]
Muğla→Montiferru	0.531 [0.495, 0.568]	0.587 [0.549, 0.624]	0.584 [0.547, 0.623]
Montiferru→Evia	0.586 [0.565, 0.606]	0.630 [0.611, 0.649]	0.624 [0.605, 0.641]
Evia→Montiferru	0.647 [0.608, 0.682]	0.568 [0.528, 0.609]	0.581 [0.539, 0.623]

degrades most of those that already transferred. Taking the better of the two adaptations per direction, 14 of the 20 end closer to chance than they began and 6 end further from it; five of those six involve Montiferru, the smallest and
575 last-added region, and move upward, while the sixth is Manavgat→Muğla moving downward from 0.470 to 0.443. The 14 to 6 split should be read at the precision of limitation (x) in Section 5.9, since Bejís→Manavgat is counted as compressed on a margin of 0.001.

In the six directions where raw transfer was below chance, the best label-
580 free method recovers at most 34 % of the gap to the within-region reference, so the remaining conditional fraction is at least 0.66 everywhere. Seven directions show *negative* recovery, meaning adaptation moves the score away from the reference; in six of those raw transfer was already above chance and adaptation destroyed that advantage. Label-free alignment therefore does
585 not act as a repair mechanism.

4.4. *Transferability diagnostics: only conditional similarity orders transfer*

Twenty candidate diagnostics from four families were each rank-correlated with the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap.

Only two diagnostics have intervals excluding zero, and both are conditional. The stronger is the sign-agreement fraction over interval-supported features, at $\rho = +0.84 [+0.58, +0.88]$. The cosine variant reaches +0.81. No marginal measure was shown to order the matrix. That includes area-of-applicability-style dissimilarity in predictor space, climatic distance
595 and geographic distance. The niche-overlap and regime families were not shown to order it either. The learned domain classifier is at ceiling, separating

Table 5: Transfer-gap decomposition (four-AOI set, 12 directions). Within = target’s within-region thermal AUC; best adapted = the better of z-score/CORAL; recovered fraction = (adapted – raw)/(within – raw), signed and unclipped, with paired bootstrap CI (1000 replicates). Montiferru directions are not part of this decomposition (per-pair absolute decompositions exist without fraction CIs). The status column asks whether the *adapted* value clears chance and uses the 2-cell adapted intervals of Table 4. The adapted arms were not recomputed at the coarser blocking of Appendix A(c), which covers the raw arm and the paired delta only.

Direction	Within	Raw	Best adapted (method)	Recovered fraction [CI]	Status
Manavgat→Bejís	0.918	0.326	0.511 (CORAL)	+0.31 [+0.28, +0.34]	recovery, chance not excluded
Bejís→Manavgat	0.870	0.444	0.555 (CORAL)	+0.26 [+0.18, +0.34]	recovery above chance
Muğla→Manavgat	0.870	0.401	0.560 (CORAL)	+0.34 [+0.28, +0.40]	recovery above chance
Bejís→Evia	0.912	0.383	0.532 (z-score)	+0.28 [+0.23, +0.32]	recovery above chance
Evia→Bejís	0.918	0.448	0.550 (z-score)	+0.22 [+0.16, +0.27]	recovery above chance
Manavgat→Muğla	0.859	0.470	0.443 (CORAL)	−0.07 [−0.12, −0.03]	negative recovery
Muğla→Bejís	0.918	0.583	0.560 (CORAL)	−0.07 [−0.16, +0.01]	negative recovery
Manavgat→Evia	0.912	0.613	0.542 (z-score)	−0.23 [−0.32, −0.14]	negative recovery
Evia→Manavgat	0.870	0.686	0.527 (CORAL)	−0.86 [−1.18, −0.60]	negative recovery
Evia→Muğla	0.859	0.577	0.530 (CORAL)	−0.17 [−0.22, −0.11]	negative recovery
Muğla→Evia	0.912	0.653	0.563 (CORAL)	−0.35 [−0.43, −0.27]	negative recovery
Bejís→Muğla	0.859	0.618	0.518 (z-score)	−0.42 [−0.51, −0.34]	negative recovery

Table 6: Transferability diagnostics versus raw thermal transfer, by family. Spearman ρ against raw transfer AUC with pair-based bootstrap 95 % CIs. The strongest member of each family in its expected direction is named. All twenty individual diagnostics are in Appendix B, Table B1.

Family	Diagnostics	Strongest member, in the expected direction	Any CI excluding 0
P(y x) conditional	6	+0.84 [+0.58, +0.88] agreement fraction, supported features	yes, 2 of 6
P(x y=1) niche overlap	5	+0.24 [−0.45, +0.74] Schoener’s D, 1-D mean	no
P(x) marginal	6	−0.32 [−0.78, +0.33] domain-classifier AUC	no
P(y) regime structure	2	+0.29 [−0.38, +0.74] log effective-N distance	no
geographic	1	−0.24 [−0.84, +0.73] centroid geodesic distance	no

source from target at $AUC \geq 0.96$ for every pair. It always succeeds, which is why it carries no ordering information.

Two limits are stated with the result rather than after it. The first is size. The index’s tie structure caps the achievable Spearman at +0.861, so the observed +0.840 sits essentially on that ceiling. Its exact one-sided permutation p is 0.0060, the smallest this tie structure can produce, against a Bonferroni threshold of 0.0026 over the nineteen computed variants. No outcome of this diagnostic could have cleared family-wise correction on ten effective pairs. The second limit is labels. Signed associations need burned labels in both regions. The family that works is therefore not available before deployment, while the family that fails is.

A third limit is about selection, and it is the sharpest of the three. The two rows that clear zero are the *supported-feature* variants. Their feature subset is chosen by whether two regions’ bootstrap intervals happen to be disjoint. That is a data-dependent selection, made on the same data, with no correction. The unselected counterparts over all nine features are $\rho = +0.50 [-0.17, +0.83]$ for the cosine and $\rho = +0.18 [-0.40, +0.72]$ for the agreement count. Both span zero. The result lives in the selection step, and is reported as such.

The families are compared on unequal samples, and equalising them does not change the ordering. The marginal, applicability, climatic and geographic rows sit on twelve directions, because those diagnostics exist only for the four-region subset. The supported-conditional rows sit on sixteen and the rest on twenty. A reader may reasonably ask whether the marginal family’s failure is a statement about power rather than about

diagnostics. Every row was therefore recomputed on the common twelve directions. The published values reproduce to 4.8×10^{-5} . The conditional rows still lead, at +0.87 [+0.65, +0.88] and +0.85 [+0.43, +0.88], and every
625 marginal row still spans zero. The ordering is not an artefact of unequal samples.

4.5. *The contrast pair: similarity is not sufficient*

The clearest single view of Table 6 needs no ranking at all. Manavgat and Muğla lie in the same country and the same fire year. They are 307
630 km apart by the centroid geodesic distance this paper uses as a diagnostic, and their nearest boundaries are 191 km apart. Their burned cells occupy the most similar environmental envelope of any pair in the matrix, with per-feature Schoener’s D of 0.77 to 0.89. Yet five of nine feature-response directions point opposite ways. Elevation is one of them, at a signed AUC of
635 0.374 [0.289, 0.471] in Manavgat against 0.611 [0.532, 0.690] in Muğla, with disjoint intervals. All four absolute thermal channels are among the others. Transfer is below chance in both directions at the point estimate, at 0.470 and 0.401. Both directions also sit inside the nominal area of applicability, at 0.875 and 0.531 of target cells.

640 Bejís and Montiferru sit at the opposite extreme. Their burned envelopes barely overlap, and they carry the most dissimilar values on every overlap measure. They transfer above chance in both directions. That half of the contrast rests on point estimates: neither direction carries a verdict at 5 km blocking, and the marginal applicability audit was never produced for
645 Montiferru. The claim is one of *sufficiency*. Similarity does not guarantee transfer, and dissimilarity does not preclude it. The claim rests on two

coexisting counterexamples, so it does not depend on the number of pairs available.

4.6. Interventions: pooling and feature removal

650 **(a) Pooled multi-region training.** Training on the pooled primary populations of the other four regions never beats the best single-source transfer for any target, and stays 0.28 to 0.50 AUC below the within-region ceiling; for two targets it falls below the pairwise mean and below chance, though only Bejís is below chance with interval support, at 0.417 [0.369, 0.467].
655 Aggregation does not manufacture the missing conditional information.

(b) Removing the direction-reversing features. The two predictors whose signed association reverses between regions are **elevation_mean** and **lst_anomaly_mean**. Retraining without them costs **−0.081** of mean within-region AUC, supported in every region (per-region deltas -0.060 , -0.130 ,
660 -0.074 , -0.063 , -0.079 , every interval entirely below zero), and changes mean transfer by **+0.014** [**−0.017**, **+0.045**], which spans zero.

 Two qualifications belong with those numbers rather than after them. First, the debit is not the thermal block’s. Dropping elevation alone accounts for -0.061 of it, from 0.888 to 0.827. Dropping the LST anomaly alone
665 accounts for -0.013 , from 0.888 to 0.875. Roughly three quarters of the cost is therefore the removal of a **baseline** terrain variable, not of a thermal channel. Second, both figures are post-selection. The two predictors were chosen because they reverse, using the same data on which the -0.081 and the $+0.014$ are then estimated. No correction for that selection is applied.

670 What this measures is a local cost with no compensating transfer gain. It does not measure an exchange, because the transfer side is a null on both

Table 7:

Configuration	Mean within-region AUC	Mean transfer AUC
full	0.888	0.541
drop <code>elevation_mean</code>	0.827	0.546
drop <code>lst_anomaly_mean</code>	0.875	0.544
drop both	0.807	0.556

arms. The thermal block’s own contribution to transfer is +0.004, with an interval spanning zero (Section 4.3). Removing the reversing predictors returns +0.014, with an interval spanning zero. Two nulls on the portability
675 axis are not a price paid.

4.7. Sensitivity analyses

Five design choices were varied with everything else held fixed. They are the Evia AOI and its prevalence, the CORAL regularisation constant, the blocking scale, the closure date of the predictor window, and the quality
680 screening of the coarse thermal input. None changes a conclusion above. Two bound how the results should be read, so they are carried into the main text here.

Coarsening the blocks from 1 km to 5 km moves the verdict counts from ten positive, seven negative and three uncertain to six, four and ten. Support
685 is removed from six verdicts and added to none. The point estimates are unchanged, but that is an identity rather than a result. The blocking scale is the resampling unit, and it cannot move an estimate computed once over all target cells.

Manavgat’s whole downstream chain was then rebuilt from a quality-

690 screened MODIS input. That changes the downscaled surface on 22,304 of
24,150 cells, by up to 10.9 °C. No signed univariate association moves by
more than +0.0003. This closes the one processing-artefact candidate for
that region’s behaviour. Appendix A reports all five arms.

4.8. The same geography, a second fire: reversal with place held constant

695 Every result above compares different places. Muğla admits a stricter test,
because a second fire burned inside the identical AOI on the identical analysis
grid. The 2021 event has a predictor window closing on 28 July. The 2022
event has a matched window closing on 20 June. Region, bounding box, cell
definition, feature registry and processing chain are the same in both arms.
700 The design holds place fixed. It does not hold season fixed, because the 2022
fire ignites about six weeks earlier, and it does not hold the population fixed,
because the 2022 arm is defined by removing the 2021 scar.

Elevation reverses with bootstrap support. In 2021 higher ground
burned preferentially, at 0.611 with its interval entirely above 0.5; in 2022
705 lower ground did, at 0.296 with its interval entirely below. The two inter-
vals are disjoint, 0.532 against 0.355, and the difference is -0.317 $[-0.414,$
 $-0.220]$. The same predictor, the same region, the same grid, and an asso-
ciation that points the opposite way in two fires eleven months apart. The
four absolute thermal channels change side as well, from bootstrap-supported
710 lower-values-burn in 2021 to higher-values-burn point estimates in 2022.

Two structural properties must be read alongside these numbers, because
they have no analogue in the twenty-direction matrix. The two arms are
not disjoint samples: they share 38,789 of the 2022 arm’s 38,790 cells, and
elevation, slope and land cover are identical to the digit across all 73,098

Table 8: Signed univariate feature-burned AUC, Muğla 2021 versus 2022. Raw AUC against **burned**, never folded to $\max(\text{AUC}, 1 - \text{AUC})$; 10-cell (≈ 5 km) spatial-block bootstrap, 1,000 replicates, seed 42 (Section 3.14). Analysis population 41,730 rows / 2,911 burned (2021) and 38,790 rows / 331 burned (2022). **Positive-carrying 5 km blocks: 70 for the 2021 arm and 11 for the 2022 arm.** Table 2’s note sets sixteen as the floor this design supports at that blocking, so the 2022 intervals here fall below the paper’s own standard and are read as indicative, exactly as the 20-cell row of Table 2 is. The 2022 arm is additionally a single compact scar, so its eleven blocks are contiguous. Read from `paper/step9g_raw/.../mugla_2021__mugla_2022_event_relative/step9g_direction_reversal_table.csv`.

Feature	2021 AUC [95 % CI]	2022 AUC [95 % CI]	Reversal
elevation_mean	0.611 [0.532, 0.690]	0.296 [0.230, 0.355]	bootstrap- supported
current_lst_mean	0.325 [0.271, 0.382]	0.515 [0.434, 0.580]	point only
current_tvdi_mean	0.336 [0.275, 0.398]	0.594 [0.475, 0.674]	point only
downscaled_lst_mean	0.307 [0.253, 0.366]	0.508 [0.435, 0.571]	point only
fused_lst_mean	0.325 [0.272, 0.383]	0.519 [0.436, 0.583]	point only
ndvi_mean	0.662 [0.616, 0.704]	0.707 [0.624, 0.777]	none
slope_mean	0.637 [0.582, 0.686]	0.558 [0.468, 0.634]	none
lst_anomaly_mean	0.485 [0.395, 0.566]	0.380 [0.249, 0.502]	none
tvdi_difference_mean	0.490 [0.396, 0.575]	0.397 [0.265, 0.514]	none

715 grid cells, so only NDVI and the six thermal channels carry new information between them. And the removal is the target’s own positive class, so in the 2022 to 2021 direction not one target positive is present in the source training population while 38,789 of 38,819 target negatives are, and membership of the source training set alone separates the 2021 target’s classes at ROC-
720 AUC 0.9996. We therefore cannot bound what that asymmetry does to the two transfer numbers, only state that a direction with this structure is not comparable to the twenty between-region directions. The elevation reversal is the finding; the structural asymmetry is a competing explanation this design cannot exclude. The known part of the bias runs the safe way: the removed
725 cells are high, with a median of 563 m, and unburned in 2022, so removing them raises the 2022 elevation AUC and makes the reversal smaller rather than larger.

4.9. What target labels cost: the recovery curve

Everything above measures a failure; this prices it. If the residual gap
730 is conditional and label-free alignment cannot close it, the missing resource is target-conditional information, and the direct way to supply it is target labels. A frozen few-shot diagnostic answers how many, for three regions across all six ordered directions, using one 10-cell (~ 5 km) spatial block as the unit of labelling effort and reading recovery against a matched target-only
735 ceiling of 0.777 to 0.824.

Thirty-two labelled blocks recover **85 to 89 % of that ceiling in three of the six directions**, two of which started below chance; 51 to 57 % in two more, the directions where the conditional gap is widest; and 30 % in the sixth. That budget is 7 to 20 % of the target’s natural-vegetation population,

roughly 800 km, so it is a real answer and not a cheap one. Two properties
of the measurement bound it and are stated in the supplement rather than
buried: at the top budget the selection pool is nearly exhausted, so the
narrow upper-budget intervals reflect saturation rather than precision, and
the labelled blocks are drawn from the event being predicted, which is not a
resource available before that event burns. The protocol, the full curve and
the limits are in Supplement S1.

5. Discussion

5.1. Principal findings

Three results carry this paper. First, the pre-fire thermal block raises
spatially blocked within-region ROC-AUC by +0.056 to +0.153 in every one
of five Mediterranean regions, with bootstrap support at 1 km and 5 km
blocking, and contributes +0.004 $[-0.028, +0.036]$ across twenty ordered
transfer directions, indistinguishable from zero and with a sign that varies
by pair. The static baseline transfers no better, at a mean of 0.537 against
0.541, so the failure is a property of this matrix rather than of the dynamic
block specifically. Removing the two reversing predictors, elevation and the
LST anomaly, costs -0.081 of within-region skill, supported in every region
and three quarters attributable to elevation, and changes transfer by +0.014
 $[-0.017, +0.045]$, which also spans zero. Second, of twenty candidate trans-
ferability diagnostics only two have intervals excluding zero and both are
conditional, while no marginal measure was shown to order the matrix, in-
cluding the area-of-applicability family that is the only one runnable before

deployment. Third, the mechanism is a reversal in the sign of the predictor-burning association, and it persists inside one study area across two fires.

765 5.2. *Why the thermal increment is real but local*

The within-region increment is not an artefact to be explained away. It replicates in five independent regions, survives coarsening of the spatial blocks to ~ 5 km with its interval intact, persists in the secondary all-valid population, and survives a predictor window closed 7 and 14 days earlier in
770 all five regions. The direction of that last result is region-specific and should not be overstated: the contribution strengthens in Bejís ($0.058 \rightarrow 0.079$ at 14 days) and Muğla ($0.115 \rightarrow 0.128$), is flat in Montiferru, and **weakens monotonically in Evia** ($0.156 \rightarrow 0.149 \rightarrow 0.135$). What holds everywhere is survival, not improvement. The transfer failure is therefore not evidence
775 that the thermal signal is spurious. It is evidence that the fitted relationship is *local*.

The natural objection is that different Mediterranean regions are simply different systems, so that a predictor meaning one thing in one place and another elsewhere is not instability but a comparison of two different systems.
780 The two Muğla events answer it directly: region, AOI, grid, feature registry and processing chain identical, static predictors identical cell by cell, and elevation’s association with burning reversing from 0.611 [$0.532, 0.690$] to 0.296 [$0.230, 0.355$], a difference of -0.317 [$-0.414, -0.220$] (Section 4.8). Holding geography fixed does not stabilise the direction of the relationship.

785 Three caveats attach and none is dismissed. Season and year are confounded, and **that confound cannot be resolved in this study area**: the events sit 42 days apart in median burn day-of-year, neither year con-

tains a second event at the other’s phase, and a calendar-matched arm would carry nine burned cells against a gate minimum of thirty. The population is
790 not held fixed either, the 2022 arm being the 2021 arm with the 2021 scar removed. And the 2022 arm rests on one compact scar and eleven positive-carrying 5 km blocks. What survives all three is the narrower reading that *something* reverses the elevation-burning association with geography, grid and pipeline held fixed, which is why Section 4.8 is reported as corroboration
795 rather than as the load-bearing evidence for the mechanism.

The sharpest supported reversal belongs to elevation, a static predictor, so instability is a property of the predictor-to-burning mapping generally rather than of thermal channels specifically. This is the same conclusion the baseline transfer arm reaches from the other direction: the instability is not
800 the thermal block’s peculiarity.

5.3. *Why the similarity-based diagnostics fail*

A diagnostic built on distance in predictor space asks one question: do the target’s predictor values look like the training data’s? That is not the question that matters when the failure is conditional. A target region can
805 sit well inside the training envelope while the relationship between those predictors and burning points the other way. Manavgat to Muğla is exactly that case, with 0.875 of target cells inside the weighted area of applicability and transfer below chance.

The same holds for the niche-overlap and regime families. At the point
810 estimates, the pair with the highest burned-niche overlap in the matrix fails in both directions while the pair with the lowest transfers in both, so no monotone function of overlap can order the outcomes. The qualifier is

not decorative: at the 5 km blocking this paper otherwise defends, neither Bejís↔Montiferru direction carries a verdict and Manavgat→Muğla loses its
815 below-chance support, so this contrast is a statement about point estimates and is made as one. The domain classifier is at ceiling for every pair, which is informative in itself: the regions are trivially separable in predictor space whether or not their transfer works, so separability carries no ordering information here.

820 5.4. *What the conditional diagnostic is, and what it is not*

The sign-agreement index reaches $\rho = +0.84$ [+0.58, +0.88] against observed transfer. Three limits are reported with it rather than after it. Its tie structure caps the achievable Spearman at +0.861, so the observed value sits essentially on its own ceiling and the ranking it supports is coarse. Its
825 exact one-sided permutation p is 0.0060, the smallest that tie structure can produce, against a Bonferroni threshold of 0.0026 over nineteen computed variants. No outcome could have cleared family-wise correction on ten effective pairs. And it needs burned labels in both regions, which makes it a mechanism diagnosis rather than a pre-deployment screen.

830 That last limit is the substantive one, and it is why the paper's practical conclusion is not "use this index instead". It is that the family of diagnostics that can be run before deployment is the family that fails here, and the information that would order transfer is the information a practitioner does not have when the decision is made. The honest statement is that these
835 twenty candidates were **not shown to order transfer**, not that they are shown incapable of it.

5.5. *What the two interventions do and do not show*

Both interventions show the same shape. Pooling four regions never beats the best single-source transfer for any target and stays 0.28 to 0.50 below
840 the within-region ceiling, so aggregation does not manufacture the missing conditional information. Removing the reversing predictors costs -0.081 of within-region skill with interval support in every region, and changes mean transfer by $+0.014$ $[-0.017, +0.045]$, which is not distinguishable from zero.

That pair of numbers is easy to read as an exchange, and it is not one.
845 The thermal block's own contribution to transfer is $+0.004$, with an interval spanning zero. Removing the reversing predictors returns $+0.014$, with an interval spanning zero. Both arms are null on the portability axis. What the interventions measure is a local cost and no compensating transfer gain. That is not a conservation law, and it is not a rate at which local skill can
850 be sold for portability. No such rate is estimated here.

5.6. *The regime hypothesis, reported as it happened*

A regime-structure explanation was stated in advance and the data confirmed the null. The regime-distance correlation has the wrong sign at the point estimate, the most regime-similar pair fails in both directions, and the
855 most regime-different pair transfers above chance. One mundane explanation can be set aside: cutting Muğla, by far the largest population, to Manavgat's cell count and separately to its positive count leaves Muğla's transfer behaviour inside the subsampling range in both roles. The error was in the hypothesised grouping, not in the data, and with ten pairs this cannot refute
860 regime typology [2] in general.

5.7. *The empirical contrast with Dimarco et al.*

Dimarco et al. [7] transfer successfully across a comparable Mediterranean design and we do not, and the two results are not in conflict. Their predictors are attributes of a place, ours describe the state of a surface in one season, 865 which suggests that the relationship between domain similarity and transfer success is predictor-class dependent.

That reading is one of at least two, and we cannot separate them here. Their response variable is human-driven **ignition**; ours is burned **area**. Ignition likelihood is dominated by human access and activity, stable and 870 near-universal drivers, whereas burned extent is dominated by spread. A predictor-class explanation and a response-variable explanation are therefore confounded in this comparison. Our own data speak against a simple predictor-class reading in any case: the static baseline transfers at a mean of 0.537 here, so within this cohort the place-attribute class does not travel 875 either.

5.8. *Implications*

For practice, transfer skill has to be *measured* rather than inferred from similarity, and the pre-deployment diagnostics currently used for that inference did not order it here. Where a model must be moved, the resource that 880 closes the gap is target labels: thirty-two labelled 5 km blocks recover 85 to 89 % of the target’s own ceiling in three of six directions, and 30 to 57 % in the rest. That is a real answer and not a cheap one, being 7 to 20 % of the target’s natural-vegetation population, and at small budgets the same intervention damages the direction that already transfers best.

885 For method development, the results point away from better unsuper-
vised alignment. Two label-free methods applied carefully moved fourteen of
twenty directions towards chance, and every direction they improved involves
the smallest region. A sign reversal is not a distribution mismatch, and no
realignment of inputs can repair it.

890 5.9. Limitations

The transfer failure is a finding, not a limitation. The limitations are the
boundaries on how far it generalises.

(i) **No meteorological covariates** enter the models, so we cannot say
how the trade-off behaves for a mixed thermal-plus-weather predictor set.

895 (ii) **Temporal transfer is measured for one region only**, and even
there year and seasonal phase are confounded, a confound that cannot be
resolved in this study area for the reason given in Section 5.2. Its 331 burned
cells also leave the thermal reversals unresolved at interval level, and the
pair holds place fixed but not population. Those two arms are addition-
900 ally the only transfer directions here computed by us rather than read from
the pipeline author’s frozen export, with his unmodified code and the same
pinned environment.

(iii) **All labels derive from a single burned-area product**, MCD64A1
[9], whose omission and commission characteristics [3] bound every model
905 evaluated here. No second burned-area product covers 2021 and 2022 at this
resolution; the companion paper reports what an independent active-fire ob-
servation says about the omission concern.

(iv) **Evia remains the most imbalance-atypical population** even
after the AOI extension, at a TSG prevalence of 0.287 against 0.038 to 0.072

910 in three of the others.

(v) **Each region contributes one fire season**, so regional concept shift is confounded with event meteorology, and distinguishing them requires multi-year labels.

(vi) **Cross-region point estimates carry an implementation tolerance** of roughly ± 0.02 to 0.03 across scikit-learn versions. All reported
915 numbers are fixed to one verified version, but exact reproduction elsewhere requires the archived environment.

(vii) **The diagnostic correlations rest on an effective sample of ten region pairs**. Both the successes and the failures of Section 4.4 should
920 be read at that power.

(viii) **Manavgat’s atypical transfer behaviour remains unexplained**.
It is where the conditional diagnosis bites hardest and where feature removal recovers most. Two candidates have now been tested and neither survives. Its predictor window was not meteorologically extreme. The second candi-
925 date was the quality-screening difference in its coarse thermal input. It was the more worrying of the two, because the change it induces correlates with elevation at $+0.615$. It does not propagate. The region’s entire downstream chain was rebuilt from a quality-screened MODIS input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9°C . No signed uni-
930 variate association moves by more than $+0.0003$. Elevation’s stays at 0.374 $[0.290, 0.472]$ in both arms, and the population is unchanged. The result is structural rather than fortunate. Elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 points of coverage. With one fire season per region the remaining

935 candidates are not separable in this design.

(ix) **The interval-support counts are less stable than the point estimates behind them.** Several verdicts sit within a thousandth of their reference value, and at 1 km blocking the published split of ten positive, seven negative and three uncertain turns on a lower bound of -0.00045 . The
940 point estimates and the sign pattern are stable; the counts are not. Every sentence in this paper that leans on an exact count of supported directions should be read at that precision.

6. Conclusions

Pre-fire thermal dryness adds a real and repeatable increment to burned-
945 area discrimination. Across five Mediterranean regions, ROC-AUC was raised by $+0.06$ to $+0.15$ over a static and near-static terrain, fuel and greenness baseline. That baseline's only time-varying member is the vegetation-index composite. The gain survived spatial blocking at about 5 km, the coarsest scale this design supports as an interval. It also survived a predictor window
950 closed up to two weeks before the first labelled burning.

That skill is local. Paired per direction, the same predictor block adds nothing distinguishable from zero to cross-region transfer. The mean is $+0.004$ over twenty directions and its interval spans zero under all four between-direction resampling units we computed. Its contribution changes
955 sign from one direction to another. The paired deltas run from -0.148 to $+0.132$, twelve positive and eight negative. At the conservative 5 km blocking, five to six directions are helped and three to four are harmed with interval support, with the rest carrying no verdict. The block is also the swing factor

at the chance line.

960 The control that fixes the meaning of all this is the baseline arm, which transfers at a mean of 0.537 against the thermal model’s 0.541. The static predictor class is not portable here either, so the honest summary is that nothing in this feature space travels, not that dynamic state is uniquely fragile. Removing the two reversing predictors, elevation and the LST anomaly, 965 costs -0.081 of mean within-region skill, with interval support in every region and roughly three quarters of it attributable to elevation, a baseline terrain variable; it changes mean transfer by $+0.014$, an estimate whose interval spans zero. A local cost is measured. No compensating transfer gain is, on either arm, so no exchange between the two is demonstrated.

970 The failure is conditional. The direction of the link between dryness and burning changes from one region to another. None of the similarity diagnostics tested here ordered the transfer matrix: predictor-space distance, domain separability, niche overlap and regime structure all failed. On ten effective pairs those null results mean not shown to order transfer, rather 975 than shown not to. It is also why label-blind adaptation compresses transfer towards chance instead of repairing it, and why pooled multi-region training does not escape it.

The practical implication is a change in what is checked before a dynamic-state fire model is transferred. The question is not whether the target region 980 lies inside the source’s environmental envelope. In our matrix, the pair with the highest envelope overlap failed in both directions, and the pair with the lowest overlap transferred in both. Both statements are made at the point estimate. The question is whether the signed feature-response directions

agree.

985 That check requires a labelled probe in the target region, and so, on
inspection, do the niche and regime diagnostics. Only the marginal family
can be computed before any target label exists, and it is the family that
failed. How large the probe has to be is not established here, so no label
budget is claimed. A signed association with a usable interval may need
990 fewer labels than a refitted model, but that was not tested. What is clear is
that the supervised recalibration that label-free alignment cannot deliver is
not cheap. Section 4.9 prices it for three regions: thirty-two labelled 5 km
blocks recovered 85 to 89 % of the target ceiling in three of six directions, 51
to 57 % in two more and 30 % in the sixth, and at its top budget the labelled
995 set already holds most of one target region’s burned cells. Where no target
labels exist, the transfer performance of such models should be treated as
unknown rather than inferred from similarity.

For model builders, local skill and portability should be reported sepa-
rately rather than assumed to travel together. A predictor block worth a
1000 large within-region increment may contribute nothing distinguishable from
zero across regions, with a sign that varies by pair, and reporting only within-
region validation hides that completely. Three extensions are left for fu-
ture work: temporal transfer, meteorological covariates, and physically nor-
malised dryness variables. Each is a step towards the self-calibrating thermal
1005 monitoring system that motivated this study.

Appendix A. Appendix A. Sensitivity analyses

Each arm below varies one design choice and leaves everything else fixed. None changes a conclusion in the main text. They are reported so that a reader can see which choices were tested, and what each was worth.

1010 **(a) Evia AOI and prevalence.** The raw transfer arms were repeated with the legacy, high-prevalence Evia box. Every qualitative conclusion is unchanged. Thermal raw transfer AUCs move by up to 0.07, the largest being Evia to Bejís at 0.378 to 0.448, and no direction changes side of the chance line.

1015 **(b) CORAL regularisation.** Over nine λ values from 0 to 10^{-1} on four directions, CORAL transfer AUC moves by at most 0.014 within any direction and 0.008 within the thermal family, so no CORAL-dependent conclusion here is sensitive to λ in that range. The companion paper reports the canonical $\lambda = 1$, which the released sweep omits.

1020 **(c) Blocking scale.** Recomputing the transfer quantities at 10-cell (≈ 5 km) blocking from the frozen per-cell predictions widens the intervals and moves the verdict counts, from ten positive, seven negative and three uncertain at 1 km to six, four and ten at 5 km. Coarser blocking therefore removes support from six verdicts and adds none.

1025 The point estimates are unchanged. That is an identity rather than a result, and it should not be offered as robustness. The blocking scale is the bootstrap *resampling unit*, and each point estimate is computed once over all target cells, so no choice of block size could have moved one. The comparison does establish two things. Every verdict that changes, changes
1030 towards "no verdict". And no direction crosses the chance line under the

widened intervals. The counts are the fragile part of this paper. The sign pattern is not thereby shown to be robust. It is simply not tested by this variation.

(d) Predictor-window closure. The predictor window was closed 7
1035 and 14 days earlier in all five regions. The thermal contribution stays positive and bootstrap-supported everywhere. The direction of the change is region-specific. It strengthens in Bejís, from 0.058 to 0.079 at 14 days, and in Muğla, from 0.115 to 0.128. It is flat in Montiferru. It weakens monotonically in Evia, from 0.156 to 0.149 to 0.135. What holds in every region is survival,
1040 not improvement. That is the claim made in Section 5.2.

(e) Quality screening of the coarse thermal input. Two of the five regions' MODIS inputs are quality-screened and three are not. The split follows export date rather than design, and it induces an elevation-correlated change at the input, at $r = +0.615$ in Manavgat. Manavgat's
1045 entire downstream chain was rebuilt from a quality-screened input. That changes the downscaled surface on 22,304 of 24,150 cells, by up to 10.9 °C. No signed univariate association moves by more than +0.0003. Elevation stays at 0.374 [0.290, 0.472] in both arms, the population is unchanged, and the within-region increment moves from [+0.055, +0.079] to [+0.054,
1050 +0.077]. The reason is structural. Elevation is a DEM variable the screening cannot touch, and fusion falls back on the MODIS-derived surface across only 2.14 points of coverage. Details are in `paper/modis_qc_downstream_propagation.md`.

Appendix B. Appendix B. All twenty transferability diagnostics

1055 Section 4.4 summarises this table by family. The full ranking is given here because the negative result is the point: a reader should be able to see every candidate that was tried, not only the families' best members, and to check that no diagnostic was dropped after it failed.

Table B1. All transferability diagnostics versus raw thermal
1060 **transfer (20 ordered directions).** Spearman ρ with pair-based bootstrap 95 % CIs. Exp. = expected sign. Rows with $n = 12$ exist only for the four-AOI subset, because those diagnostics were never produced for Montiferru. The supported-features conditional rows use the 16 directions (8 pairs) with at least one CI-supported feature.

1065 *Table note.* A null row means the diagnostic was **not shown to order transfer** on this design, not that it was shown incapable of ordering it. With ten effective region pairs the power is low, and the intervals are wide enough to admit moderate true correlations in either direction. The last row is retained rather than deleted because it was computed: on two directions
1070 the statistic has no meaningful value, and reporting that is more honest than dropping the variant.

Reading the two rows that clear zero. Both are supported-feature variants, where the predictor subset is chosen by whether two regions' bootstrap intervals happen to be disjoint. That is a data-dependent selection
1075 made on the same data, uncorrected. Their all-nine-feature counterparts are in the table and both span zero. The result lives in the selection step, and Section 4.4 says so.

Equal-sample check. The families sit on unequal samples: marginal,

Table B.9:

Diagnostic	Family	Exp.	n dir	Spearman ρ [95% CI]	CI excludes 0
Agreement fraction, supported features	$P(y x)$ conditional	+	16	+0.84 [+0.58, +0.88]	yes
Cosine, supported features	$P(y x)$ conditional	+	16	+0.81 [+0.33, +0.88]	yes
Cosine, all 9 features	$P(y x)$ conditional	+	20	+0.50 [−0.17, +0.83]	no
Vector	$P(y x)$ conditional	+	20	+0.27 [−0.36, +0.77]	no
Spearman, all 9	$P(y x)$ conditional	+	20	+0.18 [−0.40, +0.72]	no
Agreement count, all 9	$P(x y=1)$ niche	+	20	+0.24 [−0.45, +0.74]	no
Schoener’s D, 1-D mean	$P(x y=1)$ niche	+	20	+0.22 [−0.42, +0.73]	no
Warren’s I, 1-D mean	$P(x y=1)$ niche	+	20	+0.10 [−0.51, +0.68]	no
Schoener’s D, PCA-2D	$P(x y=1)$ niche	+	20	−0.07 [−0.66, +0.49]	no
Warren’s I, PCA-2D	$P(x y=1)$ niche	+	20	−0.23 [−0.75, +0.44]	no
Mahalanobis, burned centroids	$P(x)$ marginal	−	20	−0.32 [−0.78, +0.33]	no
Domain-classifier AUC	$P(x)$ marginal	−	12	−0.10 [−0.54, +0.43]	no
Predictor-space mean dissimilarity	$P(x)$ marginal	−	12	−0.08 [−0.59, +0.49]	no
Predictor-space p95 dissimilarity	$P(x)$ marginal	+	12	+0.22 [−0.48, +0.59]	no
Fraction inside weighted AoA	$P(x)$ marginal	+	12	+0.08 [−0.89, ...]	no

applicability, climatic and geographic rows on twelve directions, the supported-
 1080 conditional rows on sixteen, the rest on twenty. Recomputing every row on
 the common twelve directions reproduces the published values to 4.8×10^{-5}
 and leaves the ordering unchanged — the conditional rows still lead at +0.87
 [+0.65, +0.88] and +0.85 [+0.43, +0.88], every marginal row still spans zero.
 Source: `paper/diagnostics_common_subset.md`.

1085 **Appendix C. Appendix C. Protocol detail**

These protocols are given here in full rather than in Methods, because
 each is a specification a reader needs only when checking the corresponding
 result, and neither is needed to follow the argument. Sections 3.2, 3.3, 3.11
 and 3.14 state in summary what each does.

1090 *Appendix C.1. C.1 Burned-area label, the reconstructed analysis grid, and the admissibility gate*

Labels come from MODIS MCD64A1 Collection 6.1 (`MODIS/061/MCD64A1`)
 retrieved through Google Earth Engine [10], whose omission and commission
 characteristics [3] bound every model here. The product is exported onto
 1095 the 30 m EPSG:4326 reference grid used by the rest of the pipeline, which
 duplicates each native ~ 500 m observation across the 30 m pixels beneath
 it.

The analysis grid is therefore **reconstructed rather than native**: the
 30 m grid is partitioned into non-overlapping blocks of `round(500 / 30) =`
 1100 `17 × 17` pixels, giving a nominal cell edge of 510 m, and each cell is identified
 by its integer block indices `row_500m` and `col_500m`. This approximates the
 MODIS sinusoidal cell rather than reproducing it, being anchored to the

Landsat/EPSG:4326 reference grid, and marginal blocks are truncated. The cell is square in degrees but not on the ground, about 510 m north to south
1105 and 390 to 407 m east to west, so its area is 0.199 to 0.208 km² against a MODIS cell's 0.250 km²; every block-size label in this paper is therefore the north-south dimension. The companion paper develops what follows from that.

A cell's representative burn date is the **mode** of the positive sub-pixel
1110 day-of-year values in the block, tested against the region's label window. Because the exported raster carries no out-of-window positives, in this dataset a single in-window positive sub-pixel makes the cell burned. The fraction of positive sub-pixels agreeing with the modal date is recorded as **burn_date_pixel_agreement_fraction** but no agreement threshold is imposed. The
1115 label never affects a cell's eligibility for modelling: unburned, all-no-data and out-of-window cells all remain as the negative class.

Two safeguards are recorded rather than assumed. Cells that burned inside a region's own predictor window are removed from its analysis universe rather than counted as unburned, which ran for Muğla, Evia and Montiferru,
1120 is recorded as not run for Manavgat and has no recorded status for Bejís. Burning in earlier years is screened for no region in the five-region cohort; the only historical exclusion in the study removes the 2021 Muğla scar from the 2022 event-relative experiment of Section 3.14.

Before any modelling each region passes a gate that answers one ques-
1125 tion: of the cells labelled burned, what fraction is dominated by natural vegetation? A region is admitted as a wildfire candidate when that fraction reaches 0.50 and at least 30 burned cells are present, and is rejected as a

cropland-dominated control when the cropland fraction reaches 0.50 instead. The gate uses ESA WorldCover classes aggregated to the same reconstructed
1130 cells. Its purpose is to separate burned area produced by natural-fuel combustion from burned area produced by post-harvest stubble burning, which MCD64A1 does not distinguish. Verdicts are reported in Section 4.1.

Appendix C.2. C.2 Transferability diagnostics versus transfer

Twenty candidate diagnostics from four families are each rank-correlated
1135 against the same target quantity, the raw thermal transfer AUC over the twenty ordered directions, under one common pair-based bootstrap. The families are marginal predictor-distribution measures $P(x)$, burned-niche overlap $P(x|y=1)$, fire-regime label-pattern structure $P(y)$, and conditional feature-response direction $P(y|x)$.

1140 The **marginal** family includes area-of-applicability-style dissimilarity in scaled, importance-weighted predictor space [21, 22, 16], climatic and geographic distance, and a learned domain classifier trained to separate source from target cells. The **niche-overlap** family includes Schoener’s D [34] and Warren’s I [41] over the burned cells of each region. The **regime** family in-
1145 cludes burned-patch component counts and effective component counts. The **conditional** family is the sign-agreement index: the fraction of predictors whose signed univariate association points the same way in both regions, computed over all predictors and over the subset whose associations are individually interval-supported.

1150 Two properties of this design are stated in advance because they bound what it can show. The candidate set was fixed in a project analysis log before the correlations were computed; that log is not a formal pre-registration and

no independent timestamped registration exists. And with five regions the effective sample is ten unordered pairs, so nineteen computed variants are
1155 evaluated at that power with no family-wise error control claimed; where a diagnostic clears an interval test but not a Bonferroni threshold, both are reported.

Appendix C.3. C.3 Same-geography event-to-event comparison (Mugla 2021 versus 2022)

1160 Muğla is the one region for which a second fire event is analysed on the identical AOI and analysis grid, which allows the direction of feature-label associations to be compared with geography held fixed. The 2022 experiment is registered with windows anchored to its own event: a 58-day predictor window closing the day before ignition and a 49-day label window opening
1165 on it, matching the 2021 durations exactly.

This is not a clean temporal-transfer design and is not presented as one. The 2022 event ignites about six weeks earlier in the season than the 2021 event, so calendar year and seasonal phase are confounded and no difference can be attributed to the year alone. The registered claim is same-
1170 geography event-to-event. The pair is deliberately kept out of the twenty-direction matrix, and that exclusion is enforced in released code rather than merely asserted here.

Two further properties bound the comparison. A historical-burn exclusion applies to the 2022 arm alone, masking every cell that burned in 2021 so the
1175 previous year’s scar does not enter the following year’s analysis: it removes 3,073 cells, of which 2,941 belong to the primary population, taking it from 41,730 rows to 38,790, a drop of 7.0 %. The two arms therefore share 38,789 of

38,790 cells and three byte-identical static predictors, and in the 2022-to-2021 direction every target positive lies outside the source training population.
1180 Section 4.8 states what follows for how these two numbers may be read.

CRedit authorship contribution statement

Yunus Emre Cogurcu: Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review and editing, Supervision. **Emrehan Metin:** Software, Data curation, Investigation, Val-
1185 idation, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All satellite inputs are public and are obtained through Google Earth
1195 Engine. The processing pipeline, its configuration and the frozen numeric outputs are publicly available; the repository, the commit of record and the licence are given in the data and code availability statement of the Methods,

together with the three components that fall outside that release. No digital object identifier is minted and no archival deposit exists.

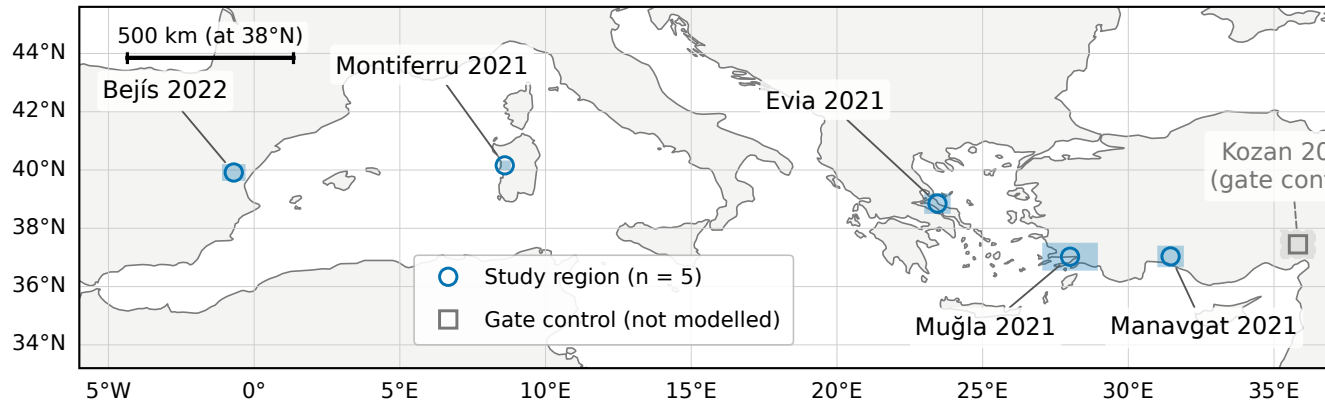


Figure C.1: **Study regions and the gate negative control.** Locations of the five wild-fire regions analysed in this study (blue, circular markers): Manavgat 2021 and Muğla 2021 (Türkiye), Bejís 2022 (Spain), North Evia 2021 (Greece, extended AOI) and Montiferru 2021 (Sardinia, Italy). Rectangles are the true analysis areas of interest; the ring marks each centroid, because at basin scale the rectangles alone are hard to locate. Kozan 2023 (grey, dashed outline, square marker) is *not* a study region: it is the negative control used to calibrate the burned-landcover admissibility gate, and it enters no model, transfer or diagnostic anywhere in this paper. Its rectangle is the true extent of its processed AOI and is larger than the study AOIs for that reason alone; the difference in size carries no methodological meaning and is not a design choice. Coastlines are Natural Earth 1:50 m; country borders are omitted deliberately. The scale bar is exact only at 38°N. Region bounding boxes match Table 1 to 5×10^{-3} degrees.

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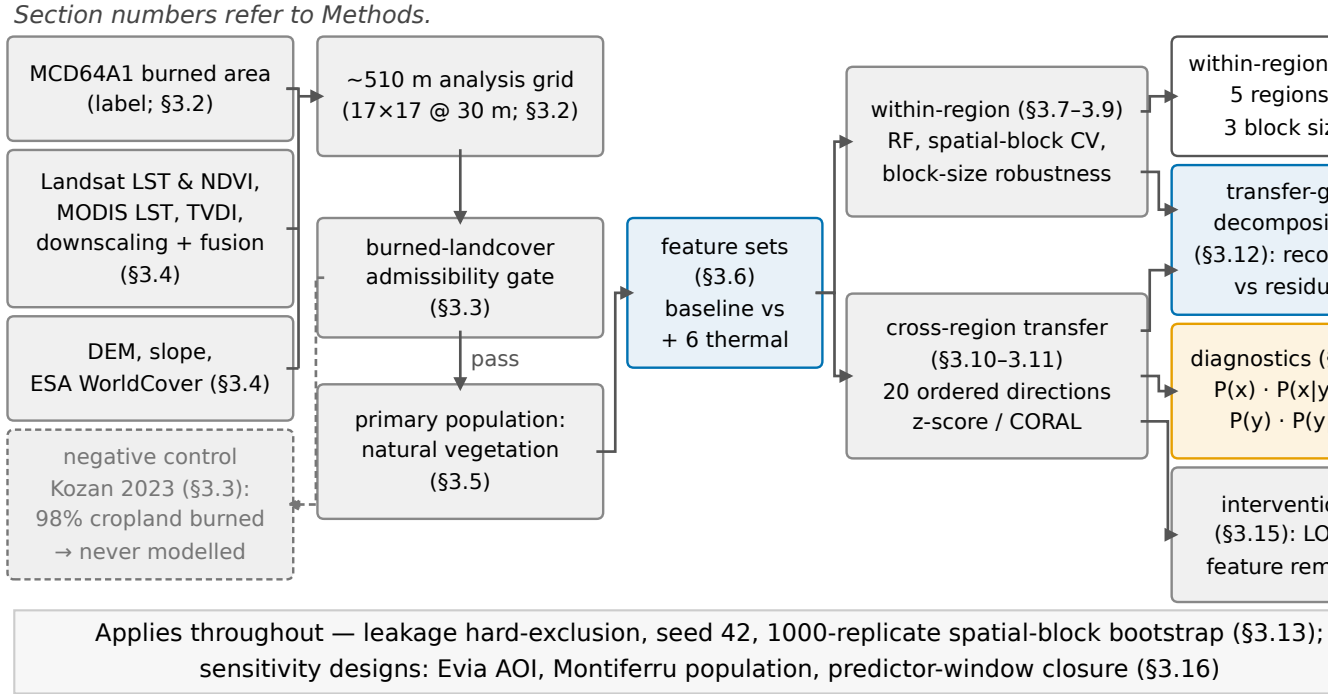


Figure C.2: Processing pipeline and evaluation programme. Flow from source products (left) to the transfer and diagnostic programme (right); every box carries the Methods subsection that specifies it. The five source products (MCD64A1 burned area, Landsat Collection 2 Level-2, MOD11A1, Copernicus DEM GLO-30 and ESA WorldCover) merge onto the reconstructed ~ 510 m analysis grid (17×17 pixels of the 30 m reference grid). The burned-landcover admissibility gate has two outcomes: regions that *pass* proceed to the natural-vegetation primary population, while the rejecting branch (grey, dashed) terminates in the Kozan 2023 negative control, whose burned area is 98% cropland and which is never modelled. The dashed connector deliberately carries no arrowhead onward, so the control cannot be read as entering the pipeline. Feature sets feed two evaluations in parallel: within-region (random forest under spatial-block cross-validation) and cross-region transfer (source-only, 20 ordered directions, with label-blind z-score and CORAL adaptation). Their levels are combined in the transfer-gap decomposition (§3.12). The footer states what applies throughout: leakage hard-exclusion, seed 42, 1000-replicate spatial-block bootstrap (§3.13), and the sensitivity designs (§3.16). Section references are verified against the Methods heading list when the figure is built.

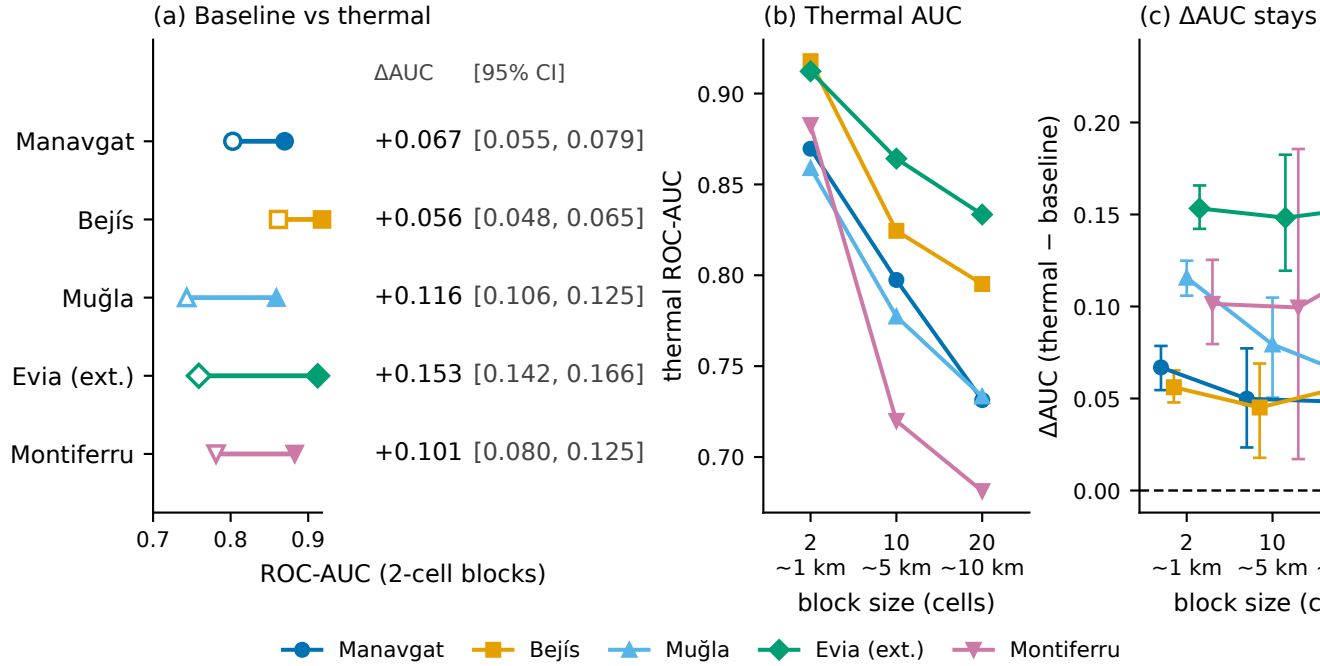


Figure C.3: **The thermal increment replicates in all five regions and survives coarser spatial blocking.** Natural-vegetation primary population; random forest with the primary weighted configuration; all intervals are 95% spatial-block bootstrap intervals (1000 replicates, seed 42). **(a)** Baseline versus thermal out-of-fold ROC-AUC per region at the default 2-cell (~ 1 km) blocking: the open marker is the static baseline feature set, the filled marker is the baseline plus the six thermal predictors, and the connecting bar is the increment. The right-hand column reports the paired ΔAUC and its 95% interval; it is a fixed column placed beyond the plotted AUC range, so no label position depends on a data value. Every interval excludes zero. **(b)** Absolute thermal ROC-AUC against spatial block size (2, 10 and 20 cells, ≈ 1 , 5 and 10 km): absolute skill falls as blocking coarsens, in every region. **(c)** The thermal increment ΔAUC against the same block sizes, with 95% intervals: the increment does *not* melt away, and its interval excludes zero in all ten of the 2- and 10-cell region-block-size combinations. The five 20-cell intervals also lie above zero but rest on 6 to 33 positive-carrying blocks each, so they are plotted as indicative rather than as intervals (Table 2, table note). In (c) the five regions are offset horizontally by 0.15 cell so their whiskers separate; **this offset is cosmetic, since all five regions are evaluated at exactly the same three block sizes** (2, 10, 20 cells), and the offset is smaller than half the spacing between block sizes, so the groups cannot merge. Colour and marker shape both encode region, so the panels remain readable in greyscale.

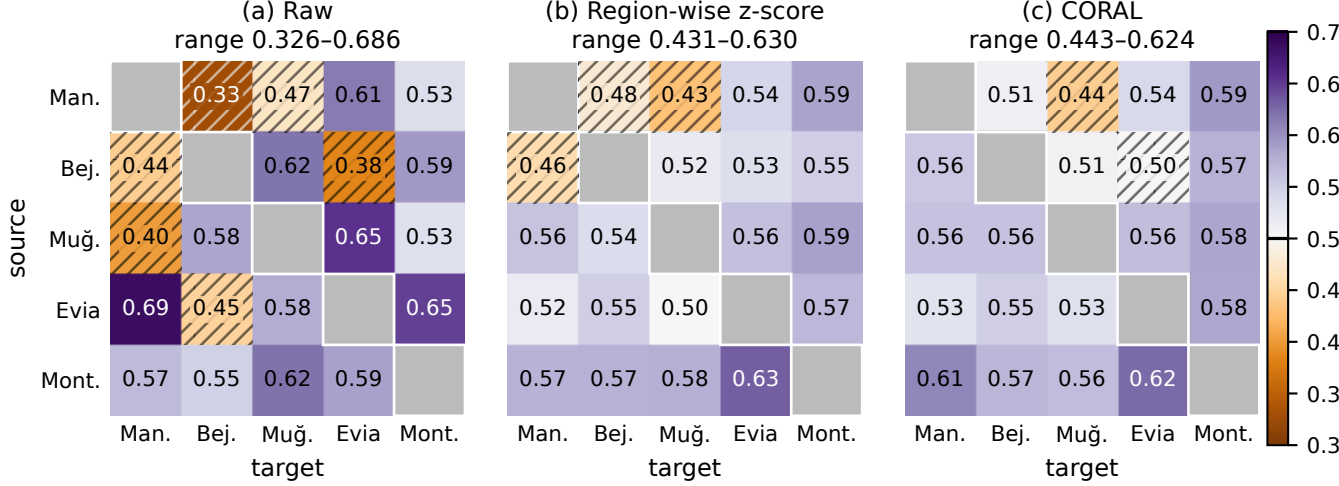


Figure C.4: **Cross-region transfer, and what label-blind adaptation does to it.** Thermal transfer ROC-AUC for all 20 ordered source–target directions among the five regions, natural-vegetation primary population. **(a)** Raw transfer, no adaptation. **(b)** After region-wise z-score standardisation. **(c)** After CORAL. Rows are the source region, columns the target; the grey diagonal is within-region performance and is *not* a transfer result. The diverging colour scale is centred exactly on chance (0.5, marked by the black rule on the colour bar), so orange is below chance and purple above. Each panel header states that panel’s range: adaptation compresses the spread from 0.326–0.686 (raw) to 0.431 to 0.630 (z-score) and 0.443 to 0.624 (CORAL). It pulls the extremes toward chance rather than lifting transfer. Cells below chance are hatched. The hatch is a redundant cue for print: a diverging palette is symmetric in luminance about its centre, so in greyscale a below-chance cell would otherwise be indistinguishable from an equally distant above-chance one. Hatching follows the *exact* value while the printed label is rounded to two decimals, so two cells print “0.50” with different hatching: CORAL Bejís→Evia is 0.4991 (hatched, below chance) and z-score Evia→Muğla is 0.5013 (un-hatched, above chance). Every cell carries its value, and every label’s colour is chosen by measured WCAG contrast against that cell’s rendered colour (worst case 4.7:1). Confidence intervals for these directions are given in Table 4.

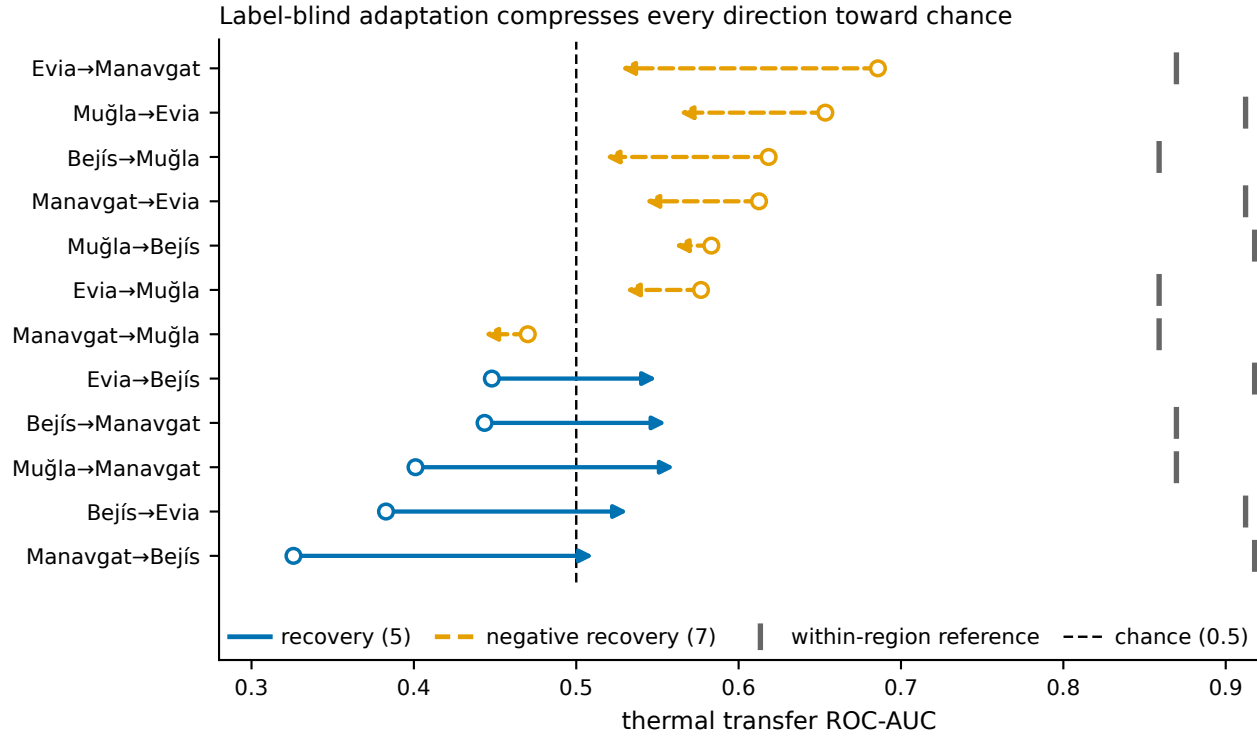


Figure C.5: **Label-blind adaptation compresses transfer toward chance rather than recovering it.** All 12 ordered directions among the four regions carrying the full decomposition (Manavgat, Bejís, Muğla, Evia), natural-vegetation primary population. Each row is one direction. The open circle is raw transfer ROC-AUC and the arrow runs to the best label-blind adapted result for that direction (region-wise z-score or CORAL, whichever is better); the grey vertical tick is the within-region reference for that target, and the dashed vertical rule is chance. **Solid blue arrows are the five directions where adaptation moves transfer toward the within-region reference (recovery); dashed orange arrows are the seven where it moves *away* from it (negative recovery).** Line style and colour both encode this class, because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. The pattern is the point: all six directions starting above chance are pushed down, and five of the six starting below chance are pushed up, so adaptation acts on the spread rather than on skill. The exception is Manavgat→Muğla, which starts below chance at 0.470 and is pushed further down to 0.443. The worst case is Evia→Manavgat, where raw transfer of 0.686 is dragged down to 0.527 against a within-region reference of 0.870, a recovered fraction of -0.862 . No arrow reaches its within-region tick.

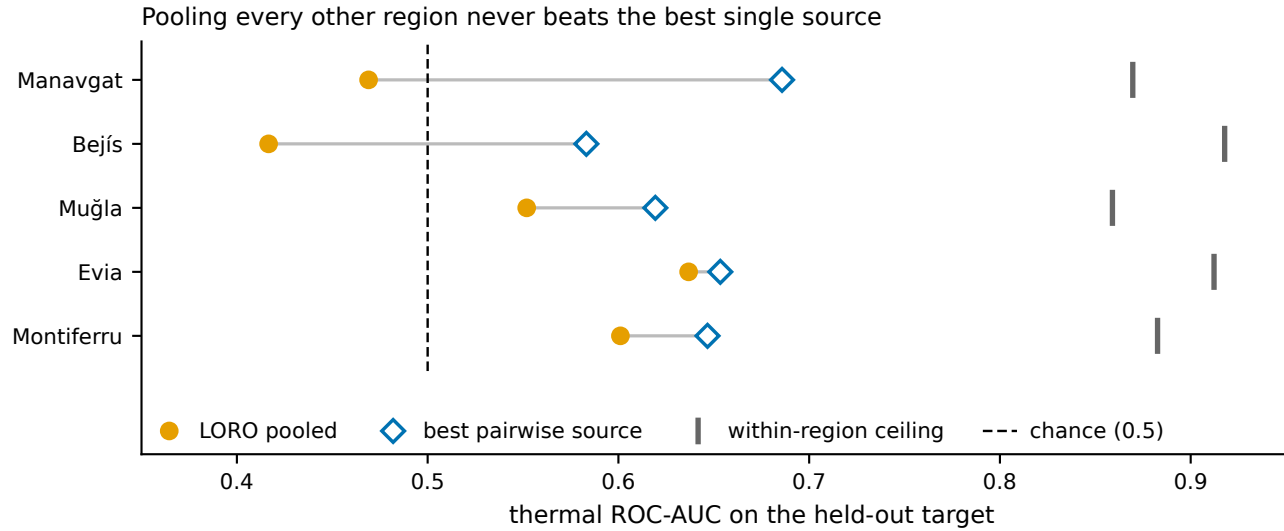


Figure C.6: **Pooling every other region never beats the best single source.** Leave-one-region-out evaluation: for each held-out target region (rows), a model trained on the pooled remaining four regions (filled orange circle) is compared against the best individual source region for that same target (open blue diamond), with the within-region ceiling as a grey vertical tick and chance as the dashed rule. Natural-vegetation primary population, thermal feature set, no adaptation. The pooled model lies to the left of the best pairwise source in all five targets without exception. Adding source regions does not accumulate transferable signal. In two targets the pooled model falls *below chance* at the point estimate (Manavgat 0.469, Bejís 0.417), and only for Bejís does the interval lie entirely below chance, while a single well-chosen source reaches 0.686 and 0.583 respectively. Every point remains far below the within-region ceiling.

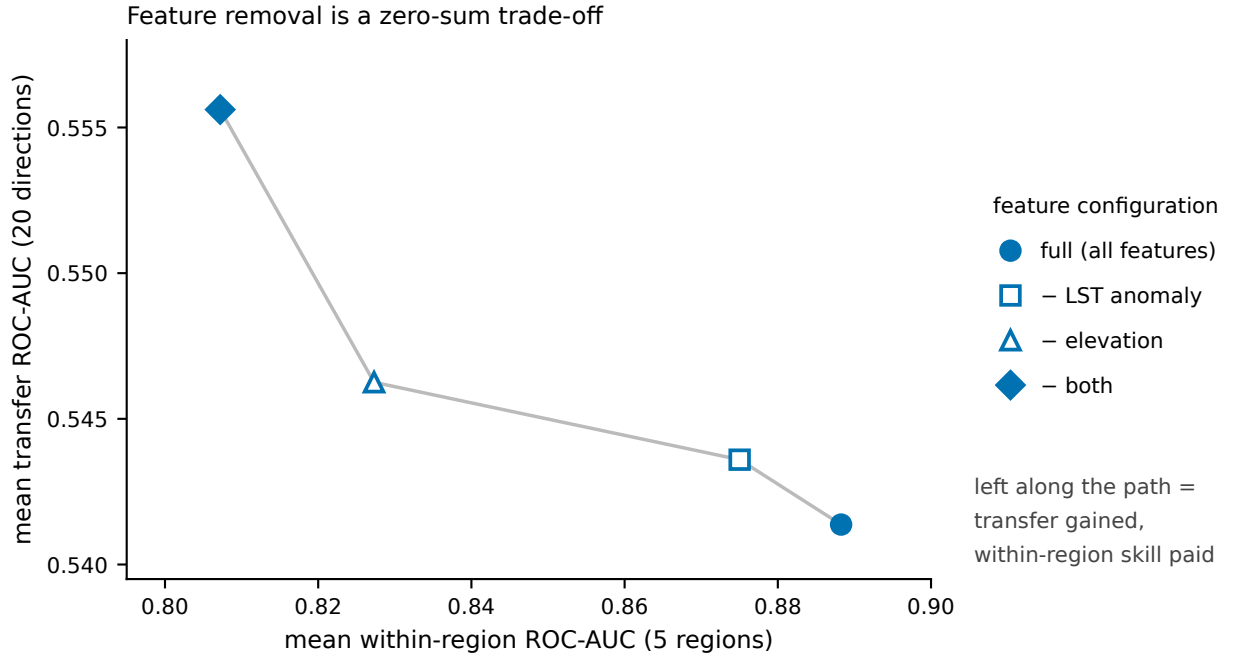


Figure C.7: **Removing the direction-reversing features is a zero-sum trade-off.**

Mean within-region ROC-AUC across the five regions (horizontal) against mean transfer ROC-AUC across the 20 ordered directions (vertical), for four feature configurations: the full thermal set, the set without the LST anomaly, without elevation, and without both. Each configuration has its own marker and is named in the key; the grey path connects them in that order. Moving left along the path, transfer rises monotonically (0.541 $\rightarrow$ 0.556) while within-region skill falls monotonically (0.888 $\rightarrow$ 0.807). Both monotonicities are verified when the figure is built, so the figure cannot silently contradict the claim. The transfer gain is 0.014 AUC and the within-region cost 0.081 AUC, about six times larger: what is removed is locally informative and non-portable, and removing it buys portability by giving up skill rather than by finding transferable signal.

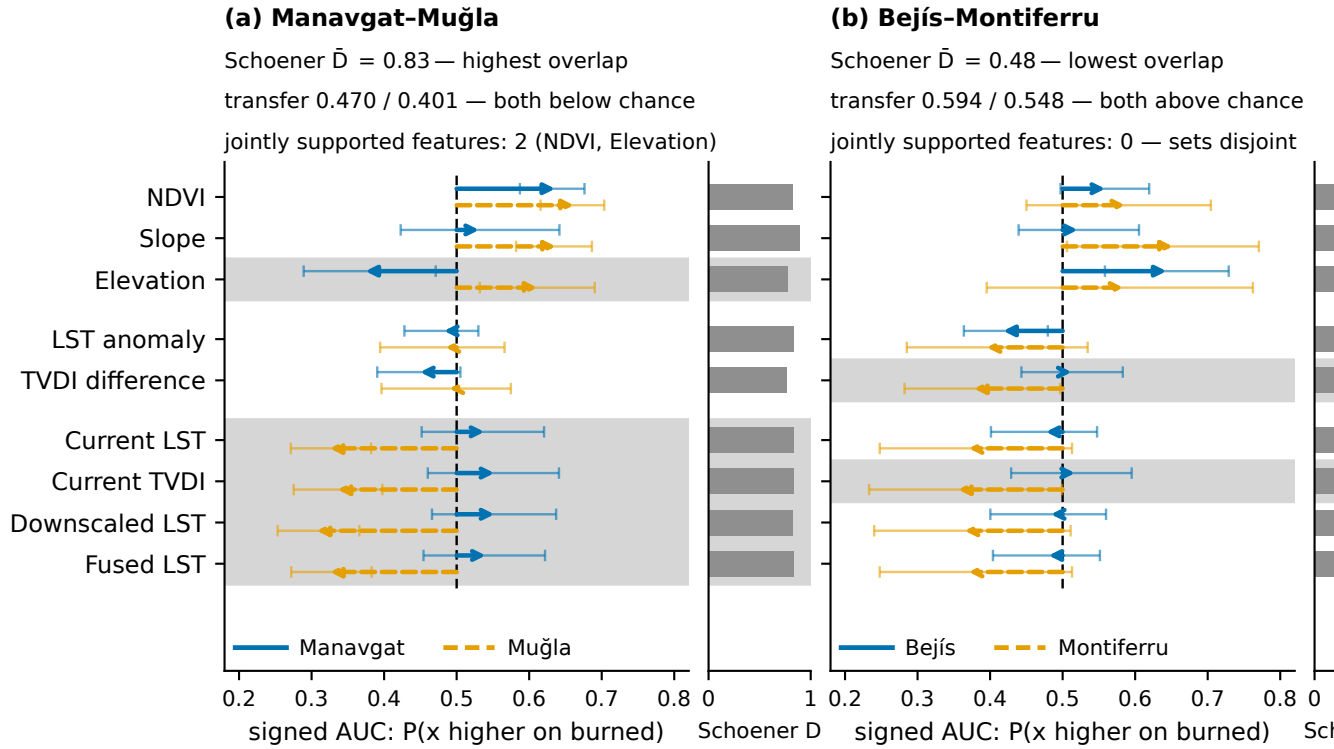


Figure C.8: **Burned-niche overlap does not determine transfer: the contrast pair.** Per-feature signed univariate AUC, the probability that a predictor takes a higher value on a burned cell than on an unburned one, for the two most informative region pairs, natural-vegetation primary population. Arrows run from chance (0.5) to the observed value, so their direction is the sign of the association and their length its strength; whiskers are 95% spatial-block (≈ 5 km) bootstrap intervals. Within each pair the two regions are distinguished by colour, by line style (solid for the first region named, dashed for the second) and by vertical offset; line style is included because the two hues differ by only 2.30:1 in relative luminance and would nearly merge in greyscale. **A filled arrowhead means that region's 95% interval excludes 0.5, so the direction is statistically supported; an open arrowhead means the interval contains 0.5 and the direction is not supported.** Rows shaded grey are features whose sign disagrees between the two regions. Grey side bars are the per-feature Schoener's D overlap. **(a)** Manavgat–Muğla, the pair with the *highest* burned-niche overlap ($\bar{D} = 0.83$), transfers below chance in both directions (0.470 and 0.401). **(b)** Bejís–Montiferru, the pair with the *lowest* overlap ($\bar{D} = 0.48$), transfers above chance in both directions (0.594 and 0.548). High niche overlap is therefore not sufficient for transfer. The arrowhead fill also exposes the coverage limit of the supported-agreement index, which is defined only on features whose direction is supported in *both* regions: there are two such features in (a) (NDVI and elevation) and **none in (b)**, where the two regions' supported sets are disjoint and Montiferru's intervals

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